Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY

Corrections and Updates Testimony

of

Gas Infrastructure and Operations Panel

Dated: June 30, 2023

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I. Introduction

Q. Please identify the members of the Gas Infrastructure and Operations Panel ("GIOP").

A. The Panel consists of Ross W. Turrini, Aaron J. Choo, Patty McVeigh, Pradheep Kileti, and Srividya Madhusudhan.

Q. Is this the same Panel that previously submitted direct testimony on behalf of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY" or the "Company") in these proceedings?

A. Yes. The defined terms in the Panel's direct testimony have the same meanings in this corrections and updates testimony.

Q. What is the purpose of the Panel's corrections and updates testimony?

A. The purpose of the Panel's testimony is to identify and explain the following corrections and updates:

- Updates to the direct capital plan – The Panel identifies the following updates and corrections to the Company's gas capital programs and projects, such as: (i) updates to the capital forecast in FY 2026 for the Jamaica Bay Line In-line Inspection ("ILI") Project; and (ii) updates to the forecasts for certain reliability projects impacting the Rate Year and Data Years.

- Updates to the indirect capital plan – The Panel describes general updates to the Company's facilities and fleet forecasts.

- Updates to the O&M forecasts – The Panel updates its O&M forecast to correct the survey component of its proposed Advanced Leak Detection ("ALD") Program, reflect the revised project scope for the EV Charging Program, and update incremental labor costs and FTEs.

Q. Is the Panel sponsoring any exhibits as part of its corrections and updates filing?

A. Yes. The Panel sponsors the following exhibits that were prepared by one or more members of the Panel or under their supervision and direction:

- Exhibit ___ (GIOP-1CU): Revised Actual and Projected Direct and Indirect Capital Expenditures: Historic Test Year, Link Period, Rate Year, and Data Years.
- Exhibit ___ (GIOP-5CU): Revised Data Sheets for Gas and Future of Heat Capital Programs.
- Exhibit ___ (GIOP-7CU): Revised Incremental O&M Non-Labor Expenditures for the Rate Year and Data Years.
- Exhibit ___ (GIOP-8CU): Revised Incremental FTE Positions by Function in the Rate Year and Data Years.

- Exhibit __ (GIOP-10CU): Revised Facilities Service Company Capital Forecast.
- Exhibit __ (GIOP-11CU): Revised Fleet Electrification Costs.

Additionally, the Company's responses to information requests ("IRs") that are relevant to the corrections and updates discussed below are included in Appendix A.

II. Corrections and Updates

A. Updates to Direct Capital Plan

Q. Please summarize the updates to the Company's direct capital investment plan and Exhibit __ (GIOP-1).

A. KEDNY's capital plan has been updated to reflect scheduling and/or budget changes in the Rate Year and Data Years for the following capital programs: (i) the Jamaica Bay ILI Project; (ii) the LNG Controls System Upgrade Project; (iii) the LNG Cyber Security Upgrade Project; (iv) the LNG Tail Gas Compressor Upgrade Project; and (v) the LNG Tank 2 Foundation Heaters Project. Each update is discussed in more detail below. The revised capital forecast is provided as Exhibit __ (GIOP-1CU).

Q. What is the impact of these changes to the Company's forecast capital expenditures in the Rate Year and Data Years?

A. The Company's updated capital forecast reflects a net decrease in the Rate Year of approximately $2.808 million, a net increase in Data Year 1 of approximately $11.800 million, a net increase in Data Year 2 of approximately $22.260 million, and a net increase in Data Year 3 of approximately $2.070 million. These figures are inclusive of cost of removal ("COR").

Table 1: Direct Capital Forecast Updates Rate Year to Data Year 3 (including COR)

($000)	FY 2024	FY 2025	FY 2026	FY 2027	FY 2028	Total
Original Forecast	$ 829,404	$ 961,556	$ 981,027	$ 953,282	$ 976,496	$ 4,701,765
Revised Forecast	$ 829,404	$ 958,749	$ 992,827	$ 975,542	$ 978,566	$ 4,735,088
Variance	$ 0	$ (2,808)	$ 11,800	$ 22,260	$ 2,070	$ 33,322

The cumulative impact of these changes over the five-year period is an increase of approximately $33.322 million to KEDNY's capital forecast. The Revenue Requirement Panel's Corrections and Updates Testimony addresses the revenue requirement impact of the Company's revised capital forecast. Additionally, the impacts of revised in-service dates for some of the projects are included in the updated revenue requirements presented in Exhibit ___ (RRP-7CU), Schedule 1.

i. Jamaica Bay Line ILI Update

Q. Please describe the changes to the Jamaica Bay Line ILI Project.

A. The Company requested funding in the Rate Year and Data Year 1 to complete the construction of this multi-year project to enable ILI on the identified segment of the transmission system. The project's scope was recently changed to include additional valve replacement and excavation work required to complete the project. The project estimate was revised to incorporate these scope changes as well as updated materials costs. The revised estimate increases the project's forecast cost in FY 2026 by $6.000 million. Additionally, the Company is correcting a typographical error in the project's data sheet. The in-service date was listed as March 2036, but should have been March 2026. A revised project data sheet is provided in Exhibit __ (GIOP-5CU). No changes are required beyond the correction to the data sheet as the Company's plant model already reflected the March 2026 in-service date.

ii. Reliability Project Updates

Q. Please describe the changes to the proposed upgrades and safety enhancement projects at the Greenpoint LNG Facility.

A. As explained in this Panel's direct testimony, the Company proposes several capital investments in the Reliability capital category to support the continued safe and reliable operation of its Greenpoint LNG Facility. Included within

these investments were projects to (i) upgrade the existing plant controls system and operator interfaces; (ii) enhance cyber security capabilities; (iii) replace the tail gas compressor and storm harden compressor assets; and (iv) complete work on LNG tank foundation heaters.

Q. Please describe the LNG – Controls System Upgrade Project.

A. The Company's capital forecast included $14.850 million and $15.858 million for the project in the Rate Year and Data Year 1, respectively, based on the original construction strategy and estimate. The construction strategy has since been revised to extend the project's duration based on National Grid's experience with similar LNG projects at the Holtsville LNG facility and allow for the continued safe operation of the facility during construction. As a result, the Company has revised the forecast project costs in the Rate Year and Data Years. The revisions decrease the project costs by $13.650 million and $9.658 million in the Rate Year and Data Year 1, respectively. The forecasts for Data Year 2 and Data Year 3 have been updated to include $8.000 million in each year based on the revised project schedule. In total, the project changes result in a decrease of $7.308 million. The revised in-service date of April 2028 has been reflected in the project's data sheet in Exhibit __ (GIOP-5CU).

Q. Please describe the changes to the LNG Cyber Security Upgrade Project.

A. The Company did not include funding for these cyber security enhancements in its initial filing because it was anticipated the project would be completed in FY 2024, prior to the Rate Year. However, the project schedule and forecast were revised to incorporate new Transportation Security Administration requirements and a recently completed security assessment. The revisions result in an increase of $1.202 million in the capital forecast for the Rate Year. The Company's response to IR No. DPS-396, provided in Appendix A, describes the updates to this project and a project data sheet is provided in Exhibit __ (GIOP-5CU).

Q. Please describe the LNG – Tail Gas Compressor Upgrade Project.

A. The Company's capital forecast included $1.319 million, $2.352 million, and $0.020 million for the Tail Gas Compressor Upgrade Project in the Rate Year, Data Year 1, and Data Year 2, respectively, based on a preliminary estimate for the project. The project has since been re-estimated based on substantial completion of the engineering design, which includes a final scope of work and associated materials costs. The re-estimate increased the Data Year 1 and Data Year 2 project costs by $5.648 million and $9.980 million, respectively, for a total project cost of $22.240 million. The revised in-service date of June 2026 has been reflected in the project's data sheet in Exhibit __ (GIOP-5CU).

Q. Please describe the LNG – Tank 2 Foundation Heaters.

A. In its April 28th filing, the Company's forecast for this project was based on a scope of work and timeline to install a new heating system for the tank by FY 2028. However, recent asset condition evaluations completed after the Company's initial filing indicated the need to replace the existing heating system sooner than originally anticipated to ensure the continued safe and reliable operation of the system. The project's execution strategy and scope have been revised to achieve a phased installation of heaters in specific zones over the next several years until the existing system is completely replaced. The Company adjusted the project's forecast to reflect the scope changes and ensure the work can be effectively managed and delivered. This adjustment moves the project's in-service date from November 2027 to June 2027. The overall impact of these modifications is an increase of $17.800 million in the project's capital forecast. The revised project forecast is included in Exhibit __ (GIOP-1CU) and summarized in Table 2, below. A revised project data sheet is provided in Exhibit __ (GIOP-5CU).

Table 2: Updates to Tank 2 Foundation Heaters Forecast (FY 2024 to Data Year 3)

Phase 1 ($000)	FY 2024	FY 2025	FY 2026	FY 2027	FY 2028	Total
Original Estimate	$ 600	$ 360	$ 190	$ 4,720	$ 9,930	$ 15,800
Revised Estimate	$ 600	$ 10,000	$ 10,000	$ 9,000	$ 4,000	$ 33,600

Q. Does the Company have any additional updates to capital projects?

A. Yes. The Company is correcting a typographical error in the Belmont Gate Station project data sheet. The in-service date was listed as November 2026, but should have been February 2030. No changes are required beyond this data sheet correction as the Company's plant model already reflected the February 2030 date. A revised project data sheet is provided in Exhibit __ (GIOP-5CU).

B. Updates to Indirect Capital Plan

Q. Please describe the updates to the indirect capital investment plan.

A. The Company has refined its fleet EV replacement schedule to better align with market availability of fit-for-purpose EVs equipped with all-wheel drive, suitable driving range, and other field and operational requirements. During the Rate and Data Years, KEDNY is now proposing to replace 196 KEDNY ICE vehicles and 18 Service Company ICE vehicles with an equivalent number of EVs, as reflected in Exhibit ___ (GIOP-11CU). Consistent with the Company's revised EV replacement schedule, the Company updated the proposed capital investment in EV chargers. Specifically, 193 Level 2 and 29 DCFC stations are now forecast, as reflected in Exhibit ___ (GIOP-1CU), as well as a reduced number of Level 2 and DCFC stations with an allocation to KEDNY. Exhibit ___ (GIOP-10CU) reflects the costs of the Level 2 and

DCFC stations allocated from the Service Company. The number of forklift chargers proposed remains unchanged. The refined forecast of EV chargers being proposed was also provided in the Company's response to IR Nos. DPS-362 and DPS-375, included in Appendix A. The revised EV charger proposal decreases the capital investment forecast in the Rate Year by $1.975 million, $0.365 million in Data Year 1, and $3.215 million in Data Year 2, and increases the forecast in by $1.020 million in Data Year 3, as shown in Exhibit __ (GIOP-1CU).

Q. Has the Company's updated the anticipated savings from the EV Program investment?

A. Yes. Based on its updated EV replacement schedule, the Company anticipates that the EV Program will provide a total savings of $0.457 million from reduced fuel costs over the Rate and Data Years, as reflected in Exhibit ___ (GIOP-11CU).

Q. Please provide any update to the Company's GHG emissions reduction benefits associated with the revised EV Program.

A. Table 3 provides the revised net emissions reductions realized from the Company's updated EV Program for KEDNY, inclusive of the allocation to KEDNY, as provided in Exhibit ___ (CLCPA-3CU). Table 3 has been updated

from the Company's initial filing to incorporate historical data specific to KEDNY's fleet instead of averages based on fleet data from all National Grid operating companies. Doing so provides a more accurate estimate of the GHG emissions benefit associated with electrifying the Company's fleet. The updated figures have also been included in the Company's revised cumulative GHG emissions reduction impact, as shown in CLCPA Panel Exhibit ___ (CLCPA-3CU).

Table 3: Updated Projected EV Program Emissions Reductions

KEDNY	Rate Year	Data Year 1	Data Year 2	Data Year 3
EV Count	34	119	132	199
Fuel Reduction (Gasoline Gallons)	21,046	74,256	81,918	123,860
Increased Load Demand (kWh)	94,569	331,588	367,361	554,188
Net Emissions Reduction (CO2e mt)	159	538	614	914

Q. Please describe the updates to the Service Company facilities programs that were described in the Panel's direct testimony.

A. Updates to the Service Company facilities projects are shown in Exhibit___ (RRP-11CU), Workpapers to Exhibit___(RRP-3CU), Schedule 9, Workpapers 7, 10, and 13. The updated EV Charger forecasts result in a

revenue requirement reduction of $0.013 million in Data Year 1, $0.066 million in Data Year 2, and $0.061 million in Data Year 3.

C. Updates to (O&M) forecasts

Q. Please describe the updates to the O&M forecasts.

A. The corrections and updates to the Company's O&M forecasts are shown in Exhibit __ (GIOP-7CU), Exhibit __ (GIOP-8CU), and Exhibit __ (GIOP-11CU). The O&M corrections and updates include changes to the ALD Program described in the Gas Safety Panel's direct testimony, the EV Charging Program, certain incremental FTEs and labor costs, and the effects of an update to inflation.

Q. Please describe the Company's updated forecast for the ALD Program.

A. The Company is updating the O&M forecast for the survey component of its ALD Program to more accurately reflect the actual volume of work that will be completed in the Rate Year and Data Years. The Company made a typographical error in the number of miles of main that would be surveyed annually to develop its initial forecast. The updated forecast is based on the current price to complete the accurate annual total – 500 miles – adjusted for inflation. The forecast updates increase non-labor O&M of $0.258 million in

the Rate Year, $0.270 million in Data Year 1, $0.283 million in Data Year 2, and $0.296 million in Data Year 3.

Q. Please describe the Company's updated forecast operating expenses for the EV program.

A. In alignment with the updated EV replacement schedule, incremental operating expenses associated with the EV program, comprising of EV and mobile EV charger lease expense, increased electricity consumption, electric infrastructure make-ready work, and allocated Service Company rent expense associated with the investment in EV charges and site backup generation, less savings realized from reduced fuel consumption, have been updated. The Company's updated EV-related O&M costs are $0.505 million in the Rate Year, $0.553 million in Data Year 1, $0.695 million in Data Year 2, and $0.682 million in Data Year 3, with additional detail provided in Exhibit ___ (GIOP-11CU), and supporting exhibits referenced therein. The refined EV replacement schedule was also provided in the Company's response to IR No. DPS-360, included in Appendix A.

Q. Please describe the Company's updates to its FTE request and labor costs.

A. Updates have been made to the hire dates for certain incremental FTEs, resulting in the removal of two FTEs. The hire date corrections have been made

in accordance with the Company's response to IR No. DPS-273. Additionally, there is a *de minimis* change in the costs for all FTEs due to a change in labor overheads as discussed in the Corrections and Updates Testimony of the Revenue Requirements Panel. These updates are reflected in Exhibit __ (GIOP-8CU) and Exhibit__(RRP-11CU), Workpapers to Exhibit (RRP-3CU), Schedule 41, Workpaper 1.

D. <u>Other Updates</u>

Q. Please describe the pending regulatory and legislative changes that may impact the Company's capital and O&M forecasts in the Rate Year and Data Years.

A. On May 4, 2023, PHMSA issued a Notice of Proposed Rulemaking (Docket No. 2021-0039) that would amend Section 113 of the PIPES Act of 2020 to increase the frequency of transmission and distribution leak surveys, require addition monitoring and surveillance, and expand the use of advanced leak detection technologies by gas utilities. The proposed changes have the potential to significantly expand the Company's leak survey activities and would require incremental resources and funding beyond what is requested in this case. Therefore, to the extent any regulations are adopted in that docket that would impact leak survey and detection requirements, the Company proposes to defer all associated incremental costs.

Q. Does the Company have any additional updates to its testimony?

A. Yes. The Company's direct testimony did not include a table outlining the minimum and proposed LPP mileage for the Rate Year and Data Years. As discussed in the Gas Safety Panel's direct testimony, the minimum mileage targets establish targets that the Company must meet annually to avoid imposition of an NRA. The proposed mileage target reflects the number of miles of LPP replacement funded in base rates, which are slightly higher than the minimum targets. The Company has provided a table below for completeness. Inclusion of this table does not impact the Company's overall LPP capital funding request or the minimum annual targets for LPP removal.

Table 4: Minimum and Proposed LPP Mileage in the Rate Year and Data Years

KEDNY	Calendar Year 2024	Calendar Year 2025	Calendar Year 2026	Calendar Year 2027	Cumulative (CY24-CY27)
Minimum Removal Targets	38	41	43	44	186
Proposed Removal Targets	43	46	48	49	186

Q. Does this conclude the Panel's testimony?

A. Yes, it does.

List of Exhibits

Exhibit ___ (GIOP-1CU): Revised Actual and Projected Direct and Indirect Capital Expenditures: Historic Test Year, Link Period, Rate Year, and Data Years.

Exhibit ___ (GIOP-5CU): Revised Data Sheets for Gas and Future of Heat Capital Programs.

Exhibit ___ (GIOP-7CU): Revised Incremental O&M Non-Labor Expenditures for the Rate Year and Data Years.

Exhibit ___ (GIOP-8CU): Revised Incremental FTE Positions by Function in the Rate Year and Data Years.

Exhibit ___ (GIOP-10CU): Revised Facilities Service Company Capital Forecast.

Exhibit ___ (GIOP-11CU): Revised Fleet Electrification Costs.

Appendix A Company Responses to Information Requests That Discuss Corrections and Updates.

Exhibit ___ (GIOP-1CU):
Revised Actual and Projected Direct and Indirect Capital Expenditures: Historic Test Year, Link Period, Rate Year, and Data Years

The Brooklyn Union Gas Company d/b/a National Grid NY
Direct Capital Expenditures (CAPEX and COR)

				Original Filing 4/28/2023					Corrections and Updates					Change				
Planning Portfolio	Classification	Investment Name	HTY (CY22)	FY24	FY25	FY26	FY27	FY28	FY24	FY25	FY26	FY27	FY28	FY24	FY25	FY26	FY27	FY28
Customer Connections	Mains & Svc	Customer Connections - Install Main	$ 14,054,337	$ 11,375,703	$ 6,935,048	$ 7,040,854	$ 6,105,880	$ 6,203,568	$ 11,375,703	$ 6,935,048	$ 7,040,854	$ 6,105,880	$ 6,203,568	$ -	$ -	$ -	$ -	$ -
Customer Connections	Mains & Svc	Customer Connections - Install Services	$ 44,505,579	$ 47,387,805	$ 28,825,360	$ 28,798,027	$ 25,471,973	$ 25,521,900	$ 47,387,805	$ 28,825,360	$ 28,798,027	$ 25,471,973	$ 25,521,900	$ -	$ -	$ -	$ -	$ -
Customer Connections	Mains & Svc	Customer Connections - Customer Contributions	$ (1,189,540)	$ (1,353,317)	$ (815,747)	$ (828,192)	$ (718,214)	$ (729,705)	$ (1,353,317)	$ (815,747)	$ (828,192)	$ (718,214)	$ (729,705)	$ -	$ -	$ -	$ -	$ -
Customer Connections	Meters/Hse Reg	Customer Connections - Install Meter/Regulator	$ 1,331,899	$ 1,221,829	$ 740,562	$ 735,279	$ 645,313	$ 641,542	$ 1,221,829	$ 740,562	$ 735,279	$ 645,313	$ 641,542	$ -	$ -	$ -	$ -	$ -
Customer Connections	Mains & Svc	Gas System Reinforcement	$ 60,102,017	$ 52,968,695	$ 32,793,298	$ 32,798,172	$ 36,713,658	$ 49,137,528	$ 52,968,695	$ 32,793,298	$ 32,798,172	$ 36,713,658	$ 49,137,528	$ -	$ -	$ -	$ -	$ -
Customer Connections	Meas. & Reg	Gas System Reinforcement - LTNY11751 - Kew Gardens Gate	$ 78,519	$ 350,000	$ 17,000,000	$ 5,300,000	$ -	$ -	$ 350,000	$ 17,000,000	$ 5,300,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Customer Connections	Meas. & Reg	Gas System Reinforcement - LTNY11986 - Flatbush Gate Station	$ -	$ -	$ -	$ 100,000	$ 250,000	$ 500,000	$ -	$ -	$ 100,000	$ 250,000	$ 500,000	$ -	$ -	$ -	$ -	$ -
Customer Connections	Meas. & Reg	Gas System Reinforcement - LTNY12025 - Belmont Gate Station	$ -	$ 1,449,000	$ 15,675,000	$ 4,000,000	$ 10,000,000	$ 15,100,000	$ 1,449,000	$ 15,675,000	$ 4,000,000	$ 10,000,000	$ 15,100,000	$ -	$ -	$ -	$ -	$ -
Customer Connections	Mains & Svc	Gas System Reinforcement - LTNY13276 - Jamaica - 175 St Transmission Main	$ -	$ -	$ -	$ -	$ -	$ 100,000	$ -	$ -	$ -	$ -	$ 100,000	$ -	$ -	$ -	$ -	$ -
Customer Connections	Mains & Svc	Gas System Reinforcement - LTNY20905 - Corona Gate Outlet Reconfigure	$ 10,701,924	$ 50,000	$ -	$ -	$ -	$ -	$ 50,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Customer Connections - Total			**$ 129,584,734**	**$ 113,449,715**	**$ 101,153,522**	**$ 77,944,138**	**$ 78,468,609**	**$ 96,474,832**	**$ 113,449,715**	**$ 101,153,522**	**$ 77,944,138**	**$ 78,468,609**	**$ 96,474,832**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**
Mandated	Mains & Svc	Corrosion	$ 180,754	$ 499,211	$ 511,292	$ 521,518	$ 531,948	$ 542,587	$ 499,211	$ 511,292	$ 521,518	$ 531,948	$ 542,587	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	CSC/Public Works - Non Reimbursable	$ 299,933,693	$ 321,725,397	$ 329,511,152	$ 335,903,668	$ 342,117,886	$ 348,549,702	$ 321,725,397	$ 329,511,152	$ 335,903,668	$ 342,117,886	$ 348,549,702	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	CSC/Public Works - Reimbursements	$ (486,058)	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	CSC/Public Works - SE853 - Flatlands Ave - Phase 2 - H30 Trans Offset Louisiana Ave & Georgia Ave	$ 1,092,846	$ 613,700	$ -	$ -	$ -	$ -	$ 613,700	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	RCV - LTNY11044 - RCV BQ1243	$ -	$ 300,000	$ 2,000,000	$ 50,000	$ -	$ -	$ 300,000	$ 2,000,000	$ 50,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	RCV - LTNY11726 - RCV BQ1238 and BQ1015	$ 78,377	$ 1,050,000	$ 8,165,000	$ 100,000	$ -	$ -	$ 1,050,000	$ 8,165,000	$ 100,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	RCV - LTNY12029 - RCV BQ1005	$ -	$ 100,000	$ 200,000	$ 4,320,000	$ -	$ -	$ 100,000	$ 200,000	$ 4,320,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	RCV - LTNY12051 - RCV BQ1019	$ -	$ -	$ 100,000	$ 200,000	$ 4,320,000	$ -	$ -	$ 100,000	$ 200,000	$ 4,320,000	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	RCV - RCV BQ1042	$ -	$ -	$ -	$ 100,000	$ 200,000	$ 2,040,000	$ -	$ -	$ 100,000	$ 200,000	$ 2,040,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	RCV - RCV BQ1059	$ -	$ -	$ -	$ 100,000	$ 200,000	$ 2,040,000	$ -	$ -	$ 100,000	$ 200,000	$ 2,040,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Integrity - Cross Bore Remediation	$ 31,321	$ 159,181	$ 100,000	$ -	$ -	$ -	$ 159,181	$ 100,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Large Diameter Pipe Rehabilitation	$ 4,033,462	$ 617,473	$ 9,437,557	$ 9,626,308	$ 9,818,834	$ 10,015,211	$ 617,473	$ 9,437,557	$ 9,626,308	$ 9,818,834	$ 10,015,211	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Large Diameter Pipe - CI Joint Encapsulation	$ -	$ -	$ 3,512,613	$ 3,580,758	$ 3,647,002	$ 3,715,566	$ -	$ 3,512,613	$ 3,580,758	$ 3,647,002	$ 3,715,566	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Large Diameter Pipe Rehabilitation - CI Main Lining - 134th St, Queens	$ 4,869,748	$ 105,000	$ -	$ -	$ -	$ -	$ 105,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Large Diameter Pipe Rehabilitation - CI Main Lining - 137th Ave	$ -	$ 4,280,000	$ 50,000	$ -	$ -	$ -	$ 4,280,000	$ 50,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Large Diameter Pipe Rehabilitation - CI Main Lining - 3rd Ave/43rd St	$ -	$ -	$ -	$ 6,337,138	$ 25,000	$ -	$ -	$ -	$ 6,337,138	$ 25,000	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Large Diameter Pipe Rehabilitation - CI Main Lining - 61st St/4thAve	$ -	$ 3,871,000	$ 50,000	$ -	$ -	$ -	$ 3,871,000	$ 50,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Large Diameter Pipe Rehabilitation - CI Main Lining - Avenue Y, Brooklyn	$ 4,422,895	$ 125,000	$ -	$ -	$ -	$ -	$ 125,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Large Diameter Pipe Rehabilitation - CI Main Lining - Prospect Pl	$ 3,990,836	$ 50,000	$ -	$ -	$ -	$ -	$ 50,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Large Diameter Pipe Rehabilitation - CISBOT	$ 6,292,066	$ 6,428,562	$ 6,848,607	$ 7,056,104	$ 7,266,787	$ 7,487,019	$ 6,428,562	$ 6,848,607	$ 7,056,104	$ 7,266,787	$ 7,487,019	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Main Protection - Access Protection Remediation	$ 30,807	$ 100,000	$ -	$ -	$ -	$ -	$ 100,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Main Replacement - Proactive - Leak Prone Pipe	$ 191,681,806	$ 156,519,058	$ 186,685,177	$ 223,395,795	$ 251,500,036	$ 269,893,511	$ 156,519,058	$ 186,685,177	$ 223,395,795	$ 251,500,036	$ 269,893,511	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Main Replacement - Reactive - Advanced Leak Detection	$ -	$ -	$ 103,770	$ 105,846	$ 107,963	$ 110,122	$ -	$ 103,770	$ 105,846	$ 107,963	$ 110,122	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Main Replacement - Reactive - Maintenance	$ 7,969,252	$ 6,857,347	$ 7,023,295	$ 7,163,761	$ 7,307,036	$ 7,453,176	$ 6,857,347	$ 7,023,295	$ 7,163,761	$ 7,307,036	$ 7,453,176	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Main Upgrades - Low Pressure Main Valve Installation	$ -	$ 1,564,500	$ -	$ -	$ -	$ -	$ 1,564,500	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Meters/Hse Reg	Meter Work - Meter Changes	$ 3,314,096	$ 3,320,970	$ 3,387,389	$ 3,455,137	$ 3,524,239	$ 3,594,724	$ 3,320,970	$ 3,387,389	$ 3,455,137	$ 3,524,239	$ 3,594,724	$ -	$ -	$ -	$ -	$ -
Mandated	Meters/Hse Reg	Meter Work - Meter Relocation	$ -		$ 2,775,000	$ 5,716,500	$ 11,775,990	$ 12,129,270		$ 2,775,000	$ 5,716,500	$ 11,775,990	$ 12,129,270	$ -	$ -	$ -	$ -	$ -
Mandated	Meters/Hse Reg	Meter Work - Proactive ERT Replacement	$ -		$ 2,172,788	$ 2,760,929	$ 3,383,818	$ 4,042,853		$ 2,172,788	$ 2,760,929	$ 3,383,818	$ 4,042,853	$ -	$ -	$ -	$ -	$ -
Mandated	Meters/Hse Reg	Automatic Meter Reading (AMR) - Replacement (ERT Purchases)	$ 3,076,810	$ 4,791,715	$ 4,470,033	$ 4,372,888	$ 4,127,896	$ 4,146,944	$ 4,791,715	$ 4,470,033	$ 4,372,888	$ 4,127,896	$ 4,146,944	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IMP - Brooklyn Backbone/Original Extension to Transco approx 42,767 feet	$ -	$ -	$ 500,000	$ 2,000,000	$ 5,000,000	$ 10,000,000	$ -	$ 500,000	$ 2,000,000	$ 5,000,000	$ 10,000,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IMP - Greenpoint LN-1.1.0 Robotic ILI approx. 11,750 feet	$ -	$ -	$ 250,000	$ 500,000	$ 7,000,000	$ 50,000	$ -	$ 250,000	$ 500,000	$ 7,000,000	$ 50,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IMP - Clove Lakes Receiver ILI	$ 2,122,702											$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IMP - Jamaica Bay Line ILI	$ 8,918,757	$ 10,311,262	$ 18,966,000	$ 9,000,000	$ -	$ -	$ 10,311,262	$ 18,966,000	$ 15,000,000	$ -	$ -	$ -	$ -	$ 6,000,000	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IMP - Southern Line Robotic ILI	$ 13,605,421	$ 15,000,000	$ 15,000,000	$ 15,000,000	$ -	$ -	$ 15,000,000	$ 15,000,000	$ 15,000,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - 4th Avenue and 6th Street	$ 632,011											$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - BBB 3rd Ave 24 inch Part R Replacement	$ 120,545	$ 500,000	$ 7,800,000	$ 7,100,000	$ 1,000,000	$ -	$ 500,000	$ 7,800,000	$ 7,100,000	$ 1,000,000	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - BBB Vanderbilt Ave 20in Part R Replacement	$ 121,908	$ 875,000	$ 2,000,000	$ 500,000	$ -	$ -	$ 875,000	$ 2,000,000	$ 500,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - BQ1014	$ 244,400	$ 7,850,000	$ -	$ -	$ -	$ -	$ 7,850,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - Brooklyn Backbone Extension to Transco Narrows	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - Clove Lakes Line AO Smith pipe replacement	$ 2,813,108	$ 19,570,000	$ 29,140,000	$ 250,000	$ -	$ -	$ 19,570,000	$ 29,140,000	$ 250,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - Coney Island Bridge Crosssing	$ -	$ 200,000	$ 200,000	$ 500,000	$ 5,000,000	$ 100,000	$ 200,000	$ 200,000	$ 500,000	$ 5,000,000	$ 100,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - Fulton & Grand Casing Work	$ 579,702	$ 554,231	$ 7,251,400	$ 58,500	$ -	$ -	$ 554,231	$ 7,251,400	$ 58,500	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - LN-1.3.0 - Coney Island	$ -	$ -	$ -	$ -	$ -	$ 200,000	$ -	$ -	$ -	$ -	$ 200,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - LN-1.4.0 - Spur to Con Ed Nar Gen	$ -	$ 200,000	$ 250,000	$ 500,000	$ 6,000,000	$ 100,000	$ 200,000	$ 250,000	$ 500,000	$ 6,000,000	$ 100,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - LN-3.0.0 - Southern Line - Part R	$ 145,839	$ 500,000	$ 750,000	$ 3,100,000	$ 4,500,000	$ 3,100,000	$ 500,000	$ 750,000	$ 3,100,000	$ 4,500,000	$ 3,100,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - LN-3.3.0 - Van Wyck/116 Ave-129 St/South Conduit	$ 8,995,243	$ 8,810	$ -	$ -	$ -	$ -	$ 8,810	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - LN-4.0.0 - Jamaica Bay Line	$ -	$ -	$ 200,000	$ 500,000	$ 5,000,000	$ -	$ -	$ 200,000	$ 500,000	$ 5,000,000	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - LN-6.1.0 - Spur to Staten Island	$ -	$ -	$ -	$ 500,000	$ 500,000	$ 2,000,000	$ -	$ -	$ 500,000	$ 500,000	$ 2,000,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - Reactive Main Replacement	$ -	$ 530,604	$ 541,000	$ 552,000	$ 563,000	$ 574,000	$ 530,604	$ 541,000	$ 552,000	$ 563,000	$ 574,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - Spur - 20" Spur to South Staten Island Gate; Tie-in, Tee, and Station Piping		$ -	$ -	$ 250,000	$ 100,000	$ 3,500,000	$ -	$ -	$ 250,000	$ 100,000	$ 3,500,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Pipeline Integrity - IVP - Verification Coney Island Spur - Replace BQ 1104	$ 209,172	$ 67,000	$ -	$ -	$ -	$ -	$ 67,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Meters/Hse Reg	Purchase Meters (New/Replacements)	$ 6,886,182	$ 7,118,040	$ 8,948,732	$ 9,156,944	$ 10,230,599	$ 11,109,465	$ 7,118,040	$ 8,948,732	$ 9,156,944	$ 10,230,599	$ 11,109,465	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Replace Pipe on Bridges	$ -		$ 650,000	$ 500,000	$ 400,000	$ 200,000		$ 650,000	$ 500,000	$ 400,000	$ 200,000	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Service Replacement - Atmospheric Corrosion Inside Inspections (Remediation)	$ 2,724,782	$ 1,895,577	$ 1,941,450	$ 1,980,279	$ 2,019,885	$ 2,060,282	$ 1,895,577	$ 1,941,450	$ 1,980,279	$ 2,019,885	$ 2,060,282	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Service Replacement - Inactive Accounts	$ 41,755	$ 417,957	$ 426,836	$ 435,273	$ 443,758	$ 452,293	$ 417,957	$ 426,836	$ 435,273	$ 443,758	$ 452,293	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Service Replacement - Reactive - Leaks	$ 5,384,544	$ 6,537,940	$ 6,696,158	$ 6,830,081	$ 6,966,683	$ 7,106,016	$ 6,537,940	$ 6,696,158	$ 6,830,081	$ 6,966,683	$ 7,106,016	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Service Replacement - Reactive - Non-Leaks - Other	$ 5,471,323	$ 6,126,947	$ 6,275,219	$ 6,400,724	$ 6,528,738	$ 6,659,313	$ 6,126,947	$ 6,275,219	$ 6,400,724	$ 6,528,738	$ 6,659,313	$ -	$ -	$ -	$ -	$ -
Mandated	Mains & Svc	Service Replacement - Valve Installation/Replacement	$ -	$ 25,000	$ 25,605	$ 26,117	$ 26,639	$ 27,172	$ 25,000	$ 25,605	$ 26,117	$ 26,639	$ 27,172	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Take Station - Over Pressure Protrection - Transco Narrows	$ -	$ 100,000	$ -	$ -	$ -	$ -	$ 100,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Bensonhurst MiniGate	$ 520,641	$ 7,399,000	$ 75,000	$ -	$ -	$ -	$ 7,399,000	$ 75,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Bush Terminal (IF-09)	$ -	$ -	$ 300,000	$ 10,000,000	$ 8,000,000	$ 200,000	$ -	$ 300,000	$ 10,000,000	$ 8,000,000	$ 200,000	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Cambria Transfer	$ -	$ -	$ -	$ -	$ 100,000	$ 200,000	$ -	$ -	$ -	$ 100,000	$ 200,000	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Canarsie Gate Station	$ -	$ -	$ -	$ -	$ -	$ 100,000	$ -	$ -	$ -	$ -	$ 100,000	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Citi Field	$ 213,026	$ 3,731,000	$ -	$ -	$ -	$ -	$ 3,731,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Clifton Gate Station	$ -	$ 560,000	$ 3,036,000	$ 70,000	$ -	$ -	$ 560,000	$ 3,036,000	$ 70,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Flatbush Mini Gate	$ -	$ -	$ -	$ -	$ -	$ 100,000	$ -	$ -	$ -	$ -	$ 100,000	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Fort Hamilton Mini Gate	$ -	$ -	$ 100,000	$ 200,000	$ 13,000,000	$ 100,000	$ -	$ 100,000	$ 200,000	$ 13,000,000	$ 100,000	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Grasmere Gate Station	$ 59,369	$ 300,000	$ 9,500,000	$ 9,170,000	$ 30,000	$ -	$ 300,000	$ 9,500,000	$ 9,170,000	$ 30,000	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Hyman station	$ 2,695,239	$ 656,000	$ -	$ -	$ -	$ -	$ 656,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Kennedy Gate	$ -	$ -	$ 100,000	$ 13,000,000	$ 150,000	$ -	$ -	$ 100,000	$ 13,000,000	$ 150,000	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Kings Plaza Mini Gate	$ 189,349	$ 175,000	$ 200,000	$ 200,000	$ 15,000,000	$ -	$ 175,000	$ 200,000	$ 200,000	$ 15,000,000	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Mariners Harbor	$ -	$ -	$ -	$ 100,000	$ 200,000	$ 13,000,000	$ -	$ -	$ 100,000	$ 200,000	$ 13,000,000	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Newtown Transfer	$ -	$ -	$ -	$ -	$ 100,000	$ 200,000	$ -	$ -	$ -	$ 100,000	$ 200,000	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - North Brooklyn Mini Gate	$ 598,285	$ 10,000	$ -	$ -	$ -	$ -	$ 10,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Mandated	Meas. & Reg	Transmission Station Integrity - PHMSA - Van Wyck Gate Station	$ -	$ -	$ -	$ 100,000	$ 200,000	$ 16,100,000	$ -	$ -	$ 100,000	$ 200,000	$ 16,100,000	$ -	$ -	$ -	$ -	$ -
Mandated - Total			**$ 593,806,014**	**$ 604,297,482**	**$ 688,227,074**	**$ 713,346,267**	**$ 747,913,738**	**$ 752,999,228**	**$ 604,297,482**	**$ 688,227,074**	**$ 719,346,267**	**$ 747,913,738**	**$ 752,999,228**	**$ -**	**$ -**	**$ 6,000,000**	**$ -**	**$ -**
Reliability	Storage	CNG - CNG Fill - Brooklyn - Canarsie - Retirement (Removal)	$ -	$ -	$ -	$ -	$ 400,000	$ -	$ -	$ -	$ -	$ 400,000	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	CNG - CNG Fill - Brooklyn - Greenpoint - Retirement (Removal)	$ -	$ -	$ -	$ -	$ 500,000	$ -	$ -	$ -	$ -	$ 500,000	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	CNG - CNG Fill - Brooklyn - Greenpoint - Station Upgrade	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	CNG - CNG Fill - Proactive	$ 349,011	$ 510,000	$ 400,000	$ 408,000	$ 200,000	$ -	$ 510,000	$ 400,000	$ 408,000	$ 200,000	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Mains & Svc	Gas Reliability	$ 7,873,298	$ 2,602,531	$ 6,546,416	$ 20,422,952	$ 9,406,206	$ 16,202,441	$ 2,602,531	$ 6,546,416	$ 20,422,952	$ 9,406,206	$ 16,202,441	$ -	$ -	$ -	$ -	$ -
Reliability	Mains & Svc	Gas Reliability - LTNY10205	$ 5,847,472	$ 1,000,000	$ -	$ -	$ -	$ -	$ 1,000,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Mains & Svc	Gas Reliability - System Automation	$ 1,186,664	$ 1,808,562	$ 1,868,245	$ 1,905,609	$ 1,943,722	$ 1,982,596	$ 1,808,562	$ 1,868,245	$ 1,905,609	$ 1,943,722	$ 1,982,596	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Gas System Control	$ -	$ 125,160	$ 129,290	$ 131,876	$ 134,514	$ 137,204	$ 125,160	$ 129,290	$ 131,876	$ 134,514	$ 137,204	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Gas System Control - Facilities Upgrades	$ -	$ -	$ -	$ 166,667	$ -	$ -	$ -	$ -	$ 166,667	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Gas System Control - MapBoard Replacement	$ -	$ 560,000	$ -	$ -	$ -	$ -	$ 560,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Gas System Control - SCADA - Upgrade/Replacement	$ -	$ -	$ 1,388,889	$ 1,388,889	$ 1,388,889	$ -	$ -	$ 1,388,889	$ 1,388,889	$ 1,388,889	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Heater Installation Program	$ 937,024	$ 1,350,000	$ 900,000	$ 750,000	$ 765,000	$ 780,300	$ 1,350,000	$ 900,000	$ 750,000	$ 765,000	$ 780,300	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Heater Installation Program - Tetco Heater #3 Replacement	$ 1,297,136	$ 4,100,000	$ 9,300,000	$ 12,000	$ -	$ -	$ 4,100,000	$ 9,300,000	$ 12,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	I&R - Reactive	$ 497,230	$ 905,721	$ 927,640	$ 946,193	$ 965,116	$ 984,419	$ 905,721	$ 927,640	$ 946,193	$ 965,116	$ 984,419	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	I&R - Training and Test Lab	$ 99,408	$ 900,000	$ -	$ -	$ -	$ -	$ 900,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -

The Brooklyn Union Gas Company d/b/a National Grid NY
Direct Capital Expenditures (CAPEX and COR)

				Original Filing 4/28/2023					Corrections and Updates					Change				
Planning Portfolio	Classification	Investment Name	HTY (CY22)	FY24	FY25	FY26	FY27	FY28	FY24	FY25	FY26	FY27	FY28	FY24	FY25	FY26	FY27	FY28
Reliability	Storage	LNG - Barge Piping Decommissioning	$ -	$ -	$ -	$ -	$ 50,000	$ 50,000	$ -	$ -	$ -	$ 50,000	$ 50,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Blanket	$ 2,086,369	$ 2,764,373	$ 2,819,660	$ 2,876,053	$ 2,933,574	$ 2,992,246	$ 2,764,373	$ 2,819,660	$ 2,876,053	$ 2,933,574	$ 2,992,246	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Boiloff Heaters/Steam Boiler Upgrade	$ 110,667	$ 175,000	$ 475,000	$ 50,000	$ 3,950,000	$ 4,275,000	$ 175,000	$ 475,000	$ 50,000	$ 3,950,000	$ 4,275,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Bulkhead Upgrade	$ 2,890	$ 100,000	$ 700,000	$ 100,000	$ 1,300,000	$ 3,900,000	$ 100,000	$ 700,000	$ 100,000	$ 1,300,000	$ 3,900,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Controls System Upgrade	$ 110,556	$ 1,038,000	$ 14,850,000	$ 15,858,000	$ -	$ -	$ 1,038,000	$ 1,200,000	$ 6,200,000	$ 8,000,000	$ 8,000,000	$ -	$ (13,650,000)	$ (9,658,000)	$ 8,000,000	$ 8,000,000
Reliability	Storage	LNG - Cyber Security Upgrade	$ -	$ 1,452,451	$ -	$ -	$ -	$ -	$ 1,452,451	$ 1,202,398	$ -	$ -	$ -	$ -	$ 1,202,398	$ -	$ -	$ -
Reliability	Storage	LNG - Dike Stabilization Tank 1 East	$ -	$ -	$ -	$ 200,000	$ 200,000	$ 350,000	$ -	$ -	$ 200,000	$ 200,000	$ 350,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Fire Protection System Upgrade	$ -	$ 500,000	$ 3,500,000	$ 5,000,000	$ 100,000	$ -	$ 500,000	$ 3,500,000	$ 5,000,000	$ 100,000	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Flare Heater Refurbishment	$ -	$ 50,000	$ 150,000	$ 350,000	$ 1,500,000	$ 3,050,000	$ 50,000	$ 150,000	$ 350,000	$ 1,500,000	$ 3,050,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Flare Refurbishment	$ -	$ 1,500,000	$ 1,500,000	$ -	$ -	$ -	$ 1,500,000	$ 1,500,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Generators Upgrade	$ 88,039	$ 350,000	$ 350,000	$ 250,000	$ 300,000	$ 1,175,000	$ 350,000	$ 350,000	$ 250,000	$ 300,000	$ 1,175,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Greenpoint Vaporization Expansion 13 & 14	$ 17,229,384	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Hydrant & Deluge Piping Upgrade	$ 641,458	$ 594,600	$ 2,683,424	$ 19,691,700	$ 6,879,890	$ -	$ 594,600	$ 2,683,424	$ 19,691,700	$ 6,879,890	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Liquefaction Mol Sieve Refurbishment	$ 616,555	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - New Control Room	$ -	$ 200,000	$ 300,000	$ 5,000,000	$ 10,000,000	$ -	$ 200,000	$ 300,000	$ 5,000,000	$ 10,000,000	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Nitrogen System Refurbishment	$ 492,818	$ 1,239,000	$ 3,000,000	$ -	$ -	$ -	$ 1,239,000	$ 3,000,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Piping Insulation Replacement & Inspection	$ 108,752	$ 1,000,000	$ 1,000,000	$ 1,000,000	$ 1,000,000	$ 1,000,000	$ 1,000,000	$ 1,000,000	$ 1,000,000	$ 1,000,000	$ 1,000,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Plant Outlet Drip Leg	$ 228,679	$ 639,000	$ 50,000	$ -	$ -	$ -	$ 639,000	$ 50,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Pump Upgrade (Tank 1)	$ 239,433	$ 1,270,000	$ 3,590,000	$ 100,000	$ -	$ -	$ 1,270,000	$ 3,590,000	$ 100,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - ReGen Heater Replacements	$ -	$ -	$ -	$ -	$ 50,000	$ 500,000	$ -	$ -	$ -	$ 50,000	$ 500,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Relocate Maintenance Area & New Control Building	$ 550,006	$ 15,582,000	$ 19,548,000	$ 12,392,000	$ -	$ -	$ 15,582,000	$ 19,548,000	$ 12,392,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - RNG Blanket	$ -	$ 200,000	$ 204,000	$ 208,080	$ 212,242	$ 216,486	$ 200,000	$ 204,000	$ 208,080	$ 212,242	$ 216,486	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Salt Water Pump House Upgrade	$ 5,253,507	$ 15,506,096	$ 25,028,870	$ 15,389,675	$ -	$ -	$ 15,506,096	$ 25,028,870	$ 15,389,675	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Security System Upgrade	$ 166,463	$ 2,000,000	$ 2,000,000	$ 2,000,000	$ -	$ -	$ 2,000,000	$ 2,000,000	$ 2,000,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Solar Panels	$ -	$ -	$ -	$ -	$ 50,000	$ 100,000	$ -	$ -	$ -	$ 50,000	$ 100,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Stormwater Drainage	$ -	$ 100,000	$ 150,000	$ 800,000	$ -	$ -	$ 100,000	$ 150,000	$ 800,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Sub M - Sub L Interconnect	$ -	$ 100,000	$ 1,100,000	$ -	$ -	$ -	$ 100,000	$ 1,100,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Tail Gas Compressor Upgrade	$ 310,996	$ 721,000	$ 1,319,000	$ 2,352,000	$ 20,000	$ -	$ 721,000	$ 1,319,000	$ 8,000,000	$ 10,000,000	$ -	$ -	$ -	$ 5,648,000	$ 9,980,000	$ -
Reliability	Storage	LNG - Tank 2 Foundation Heaters	$ 651,945	$ 600,000	$ 360,000	$ 190,000	$ 4,720,000	$ 9,930,000	$ 600,000	$ 10,000,000	$ 10,000,000	$ 9,000,000	$ 4,000,000	$ -	$ 9,640,000	$ 9,810,000	$ 4,280,000	$ (5,930,000)
Reliability	Storage	LNG - Tank 2 Upgrade	$ -	$ -	$ -	$ -	$ -	$ 500,000	$ -	$ -	$ -	$ -	$ 500,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Tank IPC Coating Upgrade	$ -	$ 3,000,000	$ 100,000	$ 750,000	$ 1,100,000	$ -	$ 3,000,000	$ 100,000	$ 750,000	$ 1,100,000	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Truck Load/Unload Station - Long Term Supply	$ 5,800,037	$ 6,490,000	$ -	$ -	$ -	$ -	$ 6,490,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Turbo Expander Generator	$ -	$ -	$ -	$ -	$ 50,000	$ 300,000	$ -	$ -	$ -	$ 50,000	$ 300,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Vapor Suppression System	$ 53,750	$ 1,290,000	$ 2,000,000	$ 1,851,000	$ 0	$ -	$ 1,290,000	$ 2,000,000	$ 1,851,000	$ 0	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Vaporizers 3 & 4 Replacement	$ 10,300,409	$ 1,610,000	$ -	$ -	$ -	$ -	$ 1,610,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Vaporizers 7 & 8 Replacement	$ -	$ 435,000	$ 3,000,000	$ 15,500,000	$ 12,600,000	$ 3,000,000	$ 435,000	$ 3,000,000	$ 15,500,000	$ 12,600,000	$ 3,000,000	$ -	$ -	$ -	$ -	$ -
Reliability	Storage	LNG - Vaporizers 9 & 10 Replacement	$ -	$ -	$ -	$ -	$ -	$ 100,000	$ -	$ -	$ -	$ -	$ 100,000	$ -	$ -	$ -	$ -	$ -
Reliability	Mains & Svc	Main Protection - Storm Hardening - Install Remote Service Shutoff Valves	$ 75,366	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ 3,741,215	$ -	$ -	$ -	$ -	$ -
Reliability	Mains & Svc	Main Upgrades - Water Intrusion	$ -	$ 150,000	$ 153,630	$ 156,703	$ 159,837	$ 163,033	$ 150,000	$ 153,630	$ 156,703	$ 159,837	$ 163,033	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - Proactive	$ 2,852,170	$ 1,500,000	$ 1,500,000	$ 1,500,000	$ 1,500,000	$ 2,958,438	$ 1,500,000	$ 1,500,000	$ 1,500,000	$ 1,500,000	$ 2,958,438	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - Citizens Gate - Bulkhead	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-03	$ 883,484	$ 5,279,000	$ 3,695,000	$ 100,000	$ -	$ -	$ 5,279,000	$ 3,695,000	$ 100,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-11	$ -	$ -	$ 106,090	$ 212,180	$ 7,000,000	$ 6,000,000	$ -	$ 106,090	$ 212,180	$ 7,000,000	$ 6,000,000	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-20	$ 177,992	$ 500,000	$ 8,076,000	$ 2,206,000	$ 20,000	$ -	$ 500,000	$ 8,076,000	$ 2,206,000	$ 20,000	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-226	$ -	$ -	$ -	$ -	$ 112,551	$ 225,102	$ -	$ -	$ -	$ 112,551	$ 225,102	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-29	$ -	$ -	$ -	$ -	$ -	$ 116,000	$ -	$ -	$ -	$ -	$ 116,000	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-303	$ 4,507,263	$ 226,000	$ -	$ -	$ -	$ -	$ 226,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-305/339	$ 1,118,046	$ 60,000	$ -	$ -	$ -	$ -	$ 60,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-54	$ 1,673,519	$ 4,430,000	$ -	$ -	$ -	$ -	$ 4,430,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-61	$ -	$ -	$ -	$ 109,000	$ 219,000	$ 13,000,000	$ -	$ -	$ 109,000	$ 219,000	$ 13,000,000	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-68 Refurbishment	$ 4,666,549	$ 25,000	$ -	$ -	$ -	$ -	$ 25,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-78	$ -	$ -	$ 106,090	$ 212,180	$ 13,000,000	$ 21,855	$ -	$ 106,090	$ 212,180	$ 13,000,000	$ 21,855	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - GOV-99	$ -	$ -	$ -	$ -	$ 112,551	$ 225,102	$ -	$ -	$ -	$ 112,551	$ 225,102	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - IF-02	$ 80,475	$ 165,000	$ 200,000	$ 13,582,000	$ 100,000	$ -	$ 165,000	$ 200,000	$ 13,582,000	$ 100,000	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Proactive Regulator Stations - IF-03	$ 103,408	$ 280,000	$ 10,000,000	$ 3,742,000	$ 160,000	$ -	$ 280,000	$ 10,000,000	$ 3,742,000	$ 160,000	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Regulator Station - Over Pressure Protection	$ 147,672	$ 280,567	$ 289,826	$ 295,622	$ 301,535	$ 307,565	$ 280,567	$ 289,826	$ 295,622	$ 301,535	$ 307,565	$ -	$ -	$ -	$ -	$ -
Reliability	Meters/Hse Reg	Smart Residential Methane Detector Program	$ -	$ 2,000,000	$ -	$ -	$ -	$ -	$ 2,000,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Transmission Mains- Greenpoint	$ 249,290	$ 535,000	$ 5,806,000	$ 50,000	$ -	$ -	$ 535,000	$ 5,806,000	$ 50,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Transmission Service Integrity	$ -	$ 500,000	$ 2,000,000	$ 3,000,000	$ 2,000,000	$ -	$ 500,000	$ 2,000,000	$ 3,000,000	$ 2,000,000	$ -	$ -	$ -	$ -	$ -	$ -
Reliability	Meas. & Reg	Tunnel Integrity Mains - Citizens Gate - Tunnel Rehab	$ 601,040	$ -	$ 2,000,000	$ 1,500,000	$ -	$ -	$ -	$ 2,000,000	$ 1,500,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Reliability- Total			**$ 80,266,229**	**$ 90,299,061**	**$ 145,171,069**	**$ 154,706,378**	**$ 87,404,625**	**$ 78,284,002**	**$ 90,299,061**	**$ 142,363,467**	**$ 160,506,378**	**$ 109,664,625**	**$ 80,354,002**	**$ -**	**$ (2,807,602)**	**$ 5,800,000**	**$ 22,260,000**	**$ 2,070,000**
Non Infrastructure	Gen Equip	Telecomm - Comm site upgrades	$ -	$ 51,420	$ 52,440	$ 53,490	$ 55,000	$ 56,000	$ 51,420	$ 52,440	$ 53,490	$ 55,000	$ 56,000	$ -	$ -	$ -	$ -	$ -
Non Infrastructure	Gen Equip	Telecomm - Radio Capital Expenditures	$ 117,133.29	$ 69,242	$ 70,918	$ 72,336	$ 73,783	$ 75,258	$ 69,242	$ 70,918	$ 72,336	$ 73,783	$ 75,258	$ -	$ -	$ -	$ -	$ -
Non Infrastructure	Gen Equip	Telecomm - Radio System Updates	$ -	$ -	$ 225,000	$ 150,000	$ 25,000	$ -	$ -	$ 225,000	$ 150,000	$ 25,000	$ -	$ -	$ -	$ -	$ -	$ -
Non Infrastructure	Gen Equip	Telecomm - Reactive	$ -	$ 49,000	$ 50,000	$ 50,000	$ 50,000	$ 50,000	$ 49,000	$ 50,000	$ 50,000	$ 50,000	$ 50,000	$ -	$ -	$ -	$ -	$ -
Non Infrastructure	Gen Equip	Tools & Equipment - Combustible gas indicator (CGI)	$ -	$ -	$ 2,675,000	$ -	$ -	$ -	$ -	$ 2,675,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Non Infrastructure	Gen Equip	Tools & Equipment - Construction	$ 5,812,913.27	$ 3,292,485	$ 3,285,280	$ 3,350,985	$ 3,418,005	$ 3,486,365	$ 3,292,485	$ 3,285,280	$ 3,350,985	$ 3,418,005	$ 3,486,365	$ -	$ -	$ -	$ -	$ -
Non Infrastructure	Gen Equip	Tools & Equipment - Field Operations	$ -	$ 1,798,547	$ 1,842,072	$ 1,878,913	$ 1,916,491	$ 1,954,821	$ 1,798,547	$ 1,842,072	$ 1,878,913	$ 1,916,491	$ 1,954,821	$ -	$ -	$ -	$ -	$ -
Non Infrastructure	Gen Equip	Tools & Equipment - I&R	$ -	$ -	$ 50,000	$ 50,000	$ 50,000	$ 50,000	$ -	$ 50,000	$ 50,000	$ 50,000	$ 50,000	$ -	$ -	$ -	$ -	$ -
Non Infrastructure	Gen Equip	Tools & Equipment - Learning & Development	$ 130,880.54	$ 187,500	$ 192,038	$ 195,763	$ 199,385	$ 203,133	$ 187,500	$ 192,038	$ 195,763	$ 199,385	$ 203,133	$ -	$ -	$ -	$ -	$ -
Non Infrastructure	Gen Equip	Tools & Equipment - Meter Testing	$ 118,436.95	$ 196,196	$ 204,904	$ 214,194	$ 224,118	$ 234,080	$ 196,196	$ 204,904	$ 214,194	$ 224,118	$ 234,080	$ -	$ -	$ -	$ -	$ -
Non-Infrastructure - Total			**$ 6,179,364**	**$ 5,644,390**	**$ 8,647,650**	**$ 6,015,682**	**$ 6,011,781**	**$ 6,109,657**	**$ 5,644,390**	**$ 8,647,650**	**$ 6,015,682**	**$ 6,011,781**	**$ 6,109,657**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**
Advanced Communication Infastructure	Meters/Hse Reg	Advanced Communication Infrastructure- AMI Network	$ -	$ -	$ -	$ -	$ 1,991,000	$ 3,484,000	$ -	$ -	$ -	$ 1,991,000	$ 3,484,000	$ -	$ -	$ -	$ -	$ -
Advanced Communication Infastructure	Meters/Hse Reg	Advanced Communication Infrastructure- Residential Methane Detectors	$ -	$ -	$ 3,305,000	$ 6,742,000	$ 10,315,000	$ 14,028,000	$ -	$ 3,305,000	$ 6,742,000	$ 10,315,000	$ 14,028,000	$ -	$ -	$ -	$ -	$ -
Advanced Communication Infastructure	Meters/Hse Reg	Advanced Communication Infrastructure- Smart Meter	$ -	$ -	$ -	$ -	$ 6,130,000	$ 9,828,000	$ -	$ -	$ -	$ 6,130,000	$ 9,828,000	$ -	$ -	$ -	$ -	$ -
Advanced Communication Infrastructure - Total			**$ -**	**$ -**	**$ 3,305,000**	**$ 6,742,000**	**$ 18,436,000**	**$ 27,340,000**	**$ -**	**$ 3,305,000**	**$ 6,742,000**	**$ 18,436,000**	**$ 27,340,000**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**
Future of Heat	FOH	Future of Heat - Gas Demand Response	$ -	$ -	$ 69,981	$ 51,776	$ 36,177	$ 13,491	$ -	$ 69,981	$ 51,776	$ 36,177	$ 13,491	$ -	$ -	$ -	$ -	$ -
Future of Heat	FOH	Future of Heat - Newtown Creek	$ 1,718,197	$ 15,713,000	$ -	$ -	$ -	$ -	$ 15,713,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Future of Heat	FOH	Future of Heat - Renewable Natural Gas (RNG) Interconnections	$ -	$ -	$ -	$ -	$ -	$ 1,000,000	$ -	$ -	$ -	$ -	$ 1,000,000	$ -	$ -	$ -	$ -	$ -
Future of Heat	FOH	Future of Heat - Renewable Natural Gas (RNG) Interconnections - #2	$ -	$ -	$ 1,080,518	$ 1,888,128	$ 895,891	$ 403,809	$ -	$ 1,080,518	$ 1,888,128	$ 895,891	$ 403,809	$ -	$ -	$ -	$ -	$ -
Future of Heat	FOH	Future of Heat - Renewable Natural Gas (RNG) Interconnections - #2 - Piping	$ -	$ -	$ -	$ 4,086,920	$ -	$ -	$ -	$ -	$ 4,086,920	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Future of Heat	FOH	Future of Heat - Renewable Natural Gas (RNG) Interconnections - #1	$ -	$ -	$ 1,080,518	$ 1,888,128	$ 895,891	$ 403,809	$ -	$ 1,080,518	$ 1,888,128	$ 895,891	$ 403,809	$ -	$ -	$ -	$ -	$ -
Future of Heat	FOH	Future of Heat - Renewable Natural Gas (RNG) Interconnections - #1 - Piping	$ -	$ -	$ -	$ 1,379,336	$ -	$ -	$ -	$ -	$ 1,379,336	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Fuutre of Heat - Total			**$ 1,718,197**	**$ 15,713,000**	**$ 2,231,017**	**$ 9,294,288**	**$ 1,827,959**	**$ 1,821,108**	**$ 15,713,000**	**$ 2,231,017**	**$ 9,294,288**	**$ 1,827,959**	**$ 1,821,108**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**
Operator Qualification Program	Gen Equip	Operator Qualification Program - Learning & Development - FTE's	$ -	$ -	$ 248,677	$ 253,501	$ 258,191	$ 263,045	$ -	$ 248,677	$ 253,501	$ 258,191	$ 263,045	$ -	$ -	$ -	$ -	$ -
Operator Qualification Program	Gen Equip	Operator Qualification Program - Learning & Development - Tools & Equipment	$ -	$ -	$ 112,500	$ 23,750	$ 24,189	$ 24,644	$ -	$ 112,500	$ 23,750	$ 24,189	$ 24,644	$ -	$ -	$ -	$ -	$ -
Operator Qualification Program	Gen Equip	Operator Qualification Program - Operations - Contract Increases	$ -	$ -	$ 12,000,000	$ 12,232,800	$ 12,459,107	$ 12,693,338	$ -	$ 12,000,000	$ 12,232,800	$ 12,459,107	$ 12,693,338	$ -	$ -	$ -	$ -	$ -
Operator Qualification Program	Gen Equip	Operator Qualification Program - Operations - FTE's	$ -	$ -	$ 72,717	$ 74,128	$ 75,499	$ 76,919	$ -	$ 72,717	$ 74,128	$ 75,499	$ 76,919	$ -	$ -	$ -	$ -	$ -
Operator Qualification Program	Gen Equip	Operator Qualification Program - Pipeline Safety - FTE's	$ -	$ -	$ 187,048	$ 190,676	$ 194,204	$ 197,855	$ -	$ 187,048	$ 190,676	$ 194,204	$ 197,855	$ -	$ -	$ -	$ -	$ -
Operator Qualification Program	Gen Equip	Operator Qualification Program - Pipeline Safety - ITS System	$ -	$ -	$ 200,000	$ 203,880	$ 207,652	$ 211,556	$ -	$ 200,000	$ 203,880	$ 207,652	$ 211,556	$ -	$ -	$ -	$ -	$ -
Operator Qualification Program - Total			**$ -**	**$ -**	**$ 12,820,942**	**$ 12,978,736**	**$ 13,218,842**	**$ 13,467,356**	**$ -**	**$ 12,820,942**	**$ 12,978,736**	**$ 13,218,842**	**$ 13,467,356**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**
KEDNY - Total - Direct			**$ 811,554,538**	**$ 829,403,649**	**$ 961,556,274**	**$ 981,027,489**	**$ 953,281,554**	**$ 976,496,185**	**$ 829,403,649**	**$ 958,748,672**	**$ 992,827,489**	**$ 975,541,554**	**$ 978,566,185**	**$ -**	**$ (2,807,602)**	**$ 11,800,000**	**$ 22,260,000**	**$ 2,070,000**
KEDNY - COR - Direct -			$ 14,484,972	**$ 23,223,302**	**$ 26,923,576**	**$ 27,468,770**	**$ 26,691,884**	**$ 27,341,893**	**$ 23,223,302**	**$ 26,844,963**	**$ 27,799,170**	**$ 27,315,164**	**$ 27,399,853**	**$ -**	**$ (78,613)**	**$ 330,400**	**$ 623,280**	**$ 57,960**
KEDNY - Total Less COR - Direct			**$ 797,069,566**	**$ 806,180,347**	**$ 934,632,698**	**$ 953,558,719**	**$ 926,589,671**	**$ 949,154,292**	**$ 806,180,347**	**$ 931,903,709**	**$ 965,028,319**	**$ 948,226,391**	**$ 951,166,332**	**$ -**	**$ (2,728,989)**	**$ 11,469,600**	**$ 21,636,720**	**$ 2,012,040**
				FY24	FY25	FY26	FY27	FY28	FY24	FY25	FY26	FY27	FY28	FY24	FY25	FY26	FY27	FY28

The Brooklyn Union Gas Company d/b/a National Grid NY
Direct Capital Expenditures (CAPEX and COR)

				Original Filing 4/28/2023					Corrections and Updates					Change				
Planning Portfolio	Classification	Investment Name	HTY (CY22)	FY24	FY25	FY26	FY27	FY28	FY24	FY25	FY26	FY27	FY28	FY24	FY25	FY26	FY27	FY28
Indirects	Indirects	Facilities - Base	$ 19,427,136	$ 4,556,737	$ 2,797,300	$ 1,392,500	$ 1,474,100	$ 1,310,000	$ 4,556,737	$ 2,797,300	$ 1,392,500	$ 1,474,100	$ 1,310,000	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - EV Charging	$ -	$ 450,000	$ 6,220,000	$ 1,500,000	$ 6,010,000	$ 2,420,000	$ 450,000	$ 4,245,000	$ 1,135,000	$ 2,795,000	$ 3,440,000	$ -	$ (1,975,000)	$ (365,000)	$ (3,215,000)	$ 1,020,000
Indirects	Indirects	Facilities - Safety	$ -	$ 2,166,460	$ 500,000	$ 500,000	$ 500,000	$ 500,000	$ 2,166,460	$ 500,000	$ 500,000	$ 500,000	$ 500,000	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Greenpoint Masterplan - Planning	$ -	$ 232,000	$ -	$ -	$ -	$ -	$ 232,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Greenpoint Warehouse Demo and Temp	$ -	$ 5,000,000	$ -	$ -	$ -	$ -	$ 5,000,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Bayridge	$ -	$ 1,957,608	$ 7,360,000	$ -	$ -	$ -	$ 1,957,608	$ 7,360,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Staten Island 200 Asset Renewal - Replacement	$ -	$ -	$ 2,700,000	$ 2,700,000	$ -	$ -	$ -	$ 2,700,000	$ 2,700,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - SI Forest Ave Asset Renewal - Replacement	$ -	$ -	$ -	$ -	$ -	$ 692,345	$ -	$ -	$ -	$ -	$ 692,345	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Jamaica DO 89-67 162nd Street Asset Renewal - Replacement	$ -	$ -	$ -	$ -	$ -	$ 1,436,050	$ -	$ -	$ -	$ -	$ 1,436,050	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Jamaica DO 127-11 Farmers Blvd Asset Renewal - Replacement	$ -	$ -	$ -	$ -	$ 843,396	$ 843,396	$ -	$ -	$ -	$ 843,396	$ 843,396	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Greenpoint Infrastructure	$ -	$ -	$ 12,125,000	$ 12,125,000	$ 12,125,000	$ 12,125,000	$ -	$ 12,125,000	$ 12,125,000	$ 12,125,000	$ 12,125,000	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Roofing Program	$ -	$ 150,000	$ 3,000,000	$ 1,000,000	$ 1,000,000	$ 1,000,000	$ 150,000	$ 3,000,000	$ 1,000,000	$ 1,000,000	$ 1,000,000	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Arc Flash Program	$ -	$ 123,075	$ 21,650	$ -	$ -	$ -	$ 123,075	$ 21,650	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - L&D Pipeline OP Qual Training Center	$ -	$ -	$ 5,000,000	$ 2,500,000	$ -	$ -	$ -	$ 5,000,000	$ 2,500,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Greenpoint Masterplan - Execution	$ -	$ 1,000,000	$ 4,600,000	$ 10,300,000	$ 17,500,000	$ 10,000,000	$ 1,000,000	$ 4,600,000	$ 10,300,000	$ 17,500,000	$ 10,000,000	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Canarsie Bathroom & Locker Reno (C087332)	$ -	$ 1,040,000	$ -	$ -	$ -	$ -	$ 1,040,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Greenpoint Bathroom & Locker Refresh (C087726)	$ -	$ 1,100,000	$ -	$ -	$ -	$ -	$ 1,100,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Canarsie Bldg 40 Exterior Windows and Façade (C085209)	$ -	$ 2,200,000	$ -	$ -	$ -	$ -	$ 2,200,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Greenpoint Site Electrical Upgrades (C081819)	$ -	$ 1,300,000	$ 1,000,000	$ -	$ -	$ -	$ 1,300,000	$ 1,000,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Greenpoint S&S Building Generator to Brightwaters (S008592)	$ -	$ 500,000	$ -	$ -	$ -	$ -	$ 500,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - Jamaica New Maintenance Facility (C088848)	$ -	$ -	$ 2,000,000	$ -	$ -	$ -	$ -	$ 2,000,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Facilities - GP - Guard Booth / Front Entrance Beautification (C089025)	$ -	$ 1,300,000	$ -	$ -	$ -	$ -	$ 1,300,000	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	IT Gas	$ 1,631,951	$ 1,934,360	$ 2,900,000	$ 2,000,000	$ 1,900,000	$ 1,900,000	$ 1,934,360	$ 2,900,000	$ 2,000,000	$ 1,900,000	$ 1,900,000	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	Fleet	$ 75,362	$ 450,000	$ 500,000	$ 500,000	$ 500,000	$ 500,000	$ 450,000	$ 500,000	$ 500,000	$ 500,000	$ 500,000	$ -	$ -	$ -	$ -	$ -
Indirects	Indirects	IMWM	$ 200,792	$ 300,000	$ 500,000	$ 500,000	$ 500,000	$ 500,000	$ 300,000	$ 500,000	$ 500,000	$ 500,000	$ 500,000	$ -	$ -	$ -	$ -	$ -
KEDNY - Total - Indirect			**$ 21,335,241**	**$ 25,760,240**	**$ 51,223,950**	**$ 35,017,500**	**$ 42,352,496**	**$ 33,226,790**	**$ 25,760,240**	**$ 49,248,950**	**$ 34,652,500**	**$ 39,137,496**	**$ 34,246,790**	**$ -**	**$ (1,975,000)**	**$ (365,000)**	**$ (3,215,000)**	**$ 1,020,000**
KEDNY - COR - Indirect -			**$ 1,066,126**	**$ 4,609,100**	**$ 1,197,300**	**$ 892,500**	**$ 974,100**	**$ 810,000**	**$ 4,609,100**	**$ 1,197,300**	**$ 892,500**	**$ 974,100**	**$ 810,000**	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**
KEDNY - Total Less COR - Indirect			**$ 20,269,115**	**$ 21,151,140**	**$ 50,026,650**	**$ 34,125,000**	**$ 41,378,396**	**$ 32,416,790**	**$ 21,151,140**	**$ 48,051,650**	**$ 33,760,000**	**$ 38,163,396**	**$ 33,436,790**	**$ -**	**$ (1,975,000)**	**$ (365,000)**	**$ (3,215,000)**	**$ 1,020,000**

KEDNY - Total - Direct & Non Direct			**$ 832,889,779**	**$ 855,163,889**	**$ 1,012,780,224**	**$ 1,016,044,989**	**$ 995,634,050**	**$ 1,009,722,975**	**$ 855,163,889**	**$ 1,007,997,622**	**$ 1,027,479,989**	**$ 1,014,679,050**	**$ 1,012,812,975**	**$ -**	**$ (4,782,602)**	**$ 11,435,000**	**$ 19,045,000**	**$ 3,090,000**
KEDNY - Total COR - Direct & Non Direct			**$ 15,551,098**	**$ 27,832,402**	**$ 28,120,876**	**$ 28,361,270**	**$ 27,665,984**	**$ 28,151,893**	**$ 27,832,402**	**$ 28,042,263**	**$ 28,691,670**	**$ 28,289,264**	**$ 28,209,853**	**$ -**	**$ (78,613)**	**$ 330,400**	**$ 623,280**	**$ 57,960**
KEDNY - Total Less COR - Direct & Non Direct			**$ 817,338,681**	**$ 827,331,487**	**$ 984,659,348**	**$ 987,683,719**	**$ 967,968,066**	**$ 981,571,082**	**$ 827,331,487**	**$ 979,955,359**	**$ 998,788,319**	**$ 986,389,786**	**$ 984,603,122**	**$ -**	**$ (4,703,989)**	**$ 11,104,600**	**$ 18,421,720**	**$ 3,032,040**

Exhibit ___ (GIOP-5CU):
Revised Data Sheets for Gas and Future of Heat Capital Programs

Gas System Reinforcement - LTNY12025 - Belmont Gate Station

Type: ☒ **Project** ☐ **Program**		**Category:** ☒ **Capital** ☐ **Both Capital & O&M**
Investment Code: 5220100507		**Region:** ☒ **KEDNY** ☐ **KEDLI**
Planning Portfolio: ☒ **Customer Connections** ☐ **Mandated** ☐ **Reliability** ☐ **Non-Infrastructure** ☐ **Indirect**		
Annual Program: ☐	**Estimated Start Date:** January 2023	**Estimated In-Service Date:** February 2030
Sponsor: Taddeo, Anthony P.		**Business Contact:** Santangelo, Amanda E.

Work Description:

The Belmont Gate Station is an existing 350 psig to 15 psig regulator station in Brownsville, Brooklyn. The station is capacity constrained. The Belmont Gate Station Project rebuilds the regulator station to increase capacity to maintain pressure above system minimum on the gas network during periods of peak demand, thereby maintaining continuous service to all gas customers. It is assumed that the project will include temperature mitigation to account for the temperature drop associated with the increased flow experiencing a pressure drop, larger intermediate and outlet piping, and a backfeed to the inlet of the station. Additionally, any integrity issues at the existing station will be addressed.

The regulator station modification is expected to be completed before the 2025/26 winter season.

Justification Summary:

Federal (49 CFR 192.623) and New York State (16 NYCRR 255.623) regulations require the Company to maintain minimum pressures on the gas system necessary to maintain reliable service to all firm customers. KEDNY's gas system is designed for a peak day with an average temperature equal to 0°F (65HDD – Heating Degree Days), with five percent of the daily send-out as a peak hour. The peak demand is based on the same forecast utilized to develop the gas supply portfolio.

The increased supply provided by the Belmont Gate Station project is required to maintain service under peak day, peak hour conditions.

Customer Benefits:

· Completion of this project will improve pressures to customers on the 15 psig system in Brownsville, Brooklyn.

The Belmont Gate Station Project will:

· Provide immediate pressure benefits to the 15 psig system by installing larger regulators and replaced undersized intermediate and outlet piping with appropriately sized piping

· Improve pressure and increase reliability by providing a backfeed to the inlet to the Belmont Gate Station which is currently undersized

· Identify and repair any integrity issues within the Belmont Gate Station facility

Funding Detail

nationalgrid

Cost Breakdown:

	FY25	FY26	FY27	FY28
CAPEX	$ 15,675,000	$ 4,000,000	$ 10,000,000	$ 15,100,000
OPEX	-	-	-	-
TOTAL	$ 15,675,000	$ 4,000,000	$ 10,000,000	$ 15,100,000

Supplemental Information

Alternatives

Alternative 1: Non-Pipe Alternative ("NPAs"). Portable CNG Injection Utilizing portable CNG injection was also considered and could temporarily provide incremental localized capacity, thereby improving system pressure. However, this option, based on a high level constructability assessment, would likely be very challenging from a siting perspective, especially considering process safety concerns, and would only be a temporary solution

Alternative 2: Do Nothing. A decision to do nothing would ultimately lead to a failure to comply with the regulations defined by Federal and New York State codes as conditions would continue to decay resulting in customer outages. In addition, restrictions on sales activities would be required in constrained areas and the Company could find itself in violation of its tariff.

The records for this station have not been evaluated, and any deficiencies found may require some level of replacement.

NPA Suitability Screening

1. **Is the project construction expected to commence and be completed within 24 months?**
   ☐ Yes ☒ No
2. **For KEDLI or NMPC, is the cost of the project less than $500k? For KEDNY, is the cost of the project less than $750k?**
   ☐ Yes ☒ No
3. **Does the pipes investment affect the critical reliability of the local or broader gas system or respond to a regulatory mandate?**
   ☒ Yes ☐ No

If questions are answered "yes," the project does not qualify for an NPA.

NPA Evaluation

Has this project been reviewed against the Company's NPA Screening and Suitability Criteria?

☒ Yes ☐ No

Please briefly explain the results: Yes, see above. Project does not qualify as it affects the critical reliability of the gas system and is needed to satisfy integrity concerns.

CLCPA/GHG Analysis

1. **Is the project/program located in an area designated as a disadvantaged community (DAC)[1]?** ☒ Yes ☐ No
2. **If so, explain how was the routing determined and how will the project impact the surrounding area?** Rebuilding of an existing Gate Station to benefit that local portion of the distribution system. Routing determined based on very specific distribution system need, constructability of below ground infrastructure, and cost. Impact of surrounding area typical to below ground utility infrastructure work.
3. **Will the project contribute to GHG emission reductions on the Company's gas network?** ☒ Yes ☐ No
4. **Please explain the project's GHG impact**. Indirect Impact. Replacing old and aging station equipment with new equipment and technologies.

Studies/References That Support the Program: N/A

Benefits
Reliability Benefits
☐ Supports growth forecast
☐ Addresses supply/capacity constraint or supply diversity needs
☐ Addresses storm/climate change resiliency
Other Customer Benefits:
☒ Improves Customer Satisfaction
☐ Reflects efficiency savings of $___
☐ Reduction in Billing Errors and Re-Bills
☐ Supports implementation of Gas Business Enablement
Safety Benefits:
☐ Enhances response time to Emergency Gas Leaks
☒ Enhances employee safety
☐ Increases automation (reduces human error)
☐ Enhances Public Safety
☐ Reduces Damages potential *(i.e., in the case of mapping system)*
☐ Addresses specific safety initiative: _____
Societal Benefits/Externalities:
☒ Emissions Reduction
☐ Reduces use of Alt. Fuel
☐ Decreases Leak Prone Pipe \Leak Backlog (Type 3 leaks)
☐ Other: _____

[1] As defined in Environmental Conservation Law Section 75-0101(5), disadvantaged communities are "communities that bear burdens of negative public health effects, environmental pollution, impacts of climate change, and possess certain socioeconomic criteria, or comprise high-concentrations of low- and moderate- income households, as identified pursuant to section 75-0111 of the [Environmental Conservation Law]."

LNG - Controls System Upgrade

Type: ☒ **Project** ☐ **Program**	**Category:** ☒ **Capital** ☐ **Both Capital & O&M**	
Investment Code: 5220000354	**Region:** ☒ **KEDNY** ☐ **KEDLI**	
Planning Portfolio: ☐ **Customer Connections** ☐ **Mandated** ☒ **Reliability** ☐ **Non-Infrastructure** ☐ **Indirect**		
Annual Program: ☐	**Estimated Start Date:** 07/29/2019	**Estimated In-Service Date:** 4/25/2028
Sponsor: Leyble, Dennis	**Business Contact:** Gray, Kevin	

Work Description:

The plant controls at the Greenpoint LNG Facility consist of various hardware and software platforms that are dispersed throughout the facility. Operators interact with these systems through PC based human machine interfaces (HMIs) or push buttons and annunciators on the main control board or local control panels within the process areas. This project intends to perform upgrades and enhancements to the existing control system and operator interfaces by implementing a new 'balance of plant' (BOP) control system that will perform monitoring, control and alarm functions various process equipment and systems throughout the plant. This BOP system will include, but not be limited to the LNG Tank System, Vaporizers, Boiloff System, Odorant System, Utilities. The liquefier contains a dedicated DCS system to manage controls and will not be upgraded as part of this project.

Justification Summary:

The project will integrate existing independent controllers and interfaces into a modern hardware/software platform that standardizes the system components when practical. This will eliminate or replace obsolete equipment and enhance system performance. These upgrades will increase the plant's reliability and enable more efficient plant operation allowing operators to easily respond to any abnormal conditions, thus increasing plant safety. Standardizing hardware will also increase the plant maintainability and allow for future upgrades to be easily integrated into the new control system. Lastly the project will further improve plant reliability by installing redundant power supplies, power circuits, and fiber optic communications.

Customer Benefits:

The primary customer benefit is the continuous, safe, and reliable supply of natural gas without unplanned outages due to facility shutdowns. Serving as the principal source of "Peak Shaving" for New York City customers, the plant also plays a strategic role in the company's gas supply portfolio. The facility has two (2) LNG storage tanks with a total design capacity of 1.6 BCF of LNG. Greenpoint LNG facility has a design output of 290 MMSCFD of natural gas. LNG peak shaving facilities operates intermittently, supplying natural gas pipelines during periods of peak need as required.

This project will add further benefits by improving on employee safety through increasing automation and implementing best design practices for human factors associated with operator interfaces (HMI) and alarm annunciation. This project will also increase the facility's reliability and resiliency through hardware standardization and redundant cabling/fiber runs.

Funding Detail

Cost Breakdown:

$000	FY25	FY26	FY27	FY28
CAPEX LNG - Controls System Upgrade	$ 1,200	$ 6,200	$ 8,000	$ 8,000
OPEX				
TOTAL	**$ 1,200**	**$ 6,200**	**$ 8,000**	**$ 8,000**

Supplemental Information

Alternatives

Alternative 1: Upgrade Control Systems, PLC System. This alternative intends to perform upgrades and enhancements to the existing control system and operator interfaces by implementing a new Balance-of-Plant (BOP) PLC system that will perform monitoring, control and alarm functions various process equipment and systems throughout the plant.

Alternative 2: Upgrade Control Systems, PLC-DCS System. This alternative intends to implement a Balance-of-Plant (BOP) with a mixed PLC-DCS system that will perform monitoring, control and alarm functions various process equipment and systems throughout the plant.

NPA Suitability Screening

1. **Is the project construction expected to commence and be completed within 24 months?**
   ☐ Yes ☒ No
2. **For KEDLI or NMPC, is the cost of the project less than $500k? For KEDNY, is the cost of the project less than $750k?**
   ☐ Yes ☒ No
3. **Does the pipes investment affect the critical reliability of the local or broader gas system or respond to a regulatory mandate?**
   ☒ Yes ☐ No

If questions are answered "yes," the project does not qualify for an NPA.

NPA Evaluation

Has this project been reviewed against the Company's NPA Screening and Suitability Criteria?

☒ Yes ☐ No

Please briefly explain the results: Project is not suitable for an NPA as it affects the critical reliability of the gas system and mitigates integrity concerns. No NPA, or combination of NPAs, would remove the need for this project at this time.

CLCPA/GHG Analysis

1. **Is the project/program located in an area designated as a disadvantaged community (DAC)[1]?** ☒ Yes ☐ No
2. **If so, explain how was the routing determined and how will the project impact the surrounding area?** Project is within National Grid's property. Minimal impacts to community.
3. **Will the project contribute to GHG emission reductions on the Company's gas network?** ☐ Yes ☒ No
4. **Please explain the project's GHG impact.** No Material Impact. This project is required for reliability and is not anticipated to result in GHG emissions reductions.

Studies/References That Support the Program:

Benefits
Reliability Benefits
☐ Supports growth forecast
☐ Addresses supply/capacity constraint or supply diversity needs
☐ Addresses storm/climate change resiliency
Other Customer Benefits:
☐ Improves Customer Satisfaction
☐ Reflects efficiency savings of $___
☐ Reduction in Billing Errors and Re-Bills
☐ Supports implementation of Gas Business Enablement
Safety Benefits:
☐ Enhances response time to Emergency Gas Leaks
☒ Enhances employee safety
☒ Increases automation (reduces human error)
☐ Enhances Public Safety
☐ Reduces Damages potential *(i.e., in the case of mapping system)*
☐ Addresses specific safety initiative: _____
Societal Benefits/Externalities:
☐ Emissions Reduction
☐ Reduces use of Alt. Fuel
☐ Decreases Leak Prone Pipe \Leak Backlog (Type 3 leaks)
☐ Other: _____

[1] As defined in Environmental Conservation Law Section 75-0101(5), disadvantaged communities are "communities that bear burdens of negative public health effects, environmental pollution, impacts of climate change, and possess certain socioeconomic criteria, or comprise high-concentrations of low- and moderate- income households, as identified pursuant to section 75-0111 of the [Environmental Conservation Law]."

LNG - Cyber Security Upgrade

Type: ☒ **Project** ☐ **Program**		**Category:** ☒ **Capital** ☐ **Both Capital & O&M**
Investment Code: 5220101837		**Region:** ☒ **KEDNY** ☐ **KEDLI**
Planning Portfolio: ☐ **Customer Connections** ☐ **Mandated** ☒ **Reliability** ☐ **Non-Infrastructure** ☐ **Indirect**		
Annual Program: ☐	**Estimated Start Date:** 04/01/2024	**Estimated In-Service Date:** 03/31/2025
Sponsor: Leyble, Dennis		**Business Contact:** Gray, Kevin

Work Description:

Through this project, National Grid's Information Technology (NG IT) organization will enhance cybersecurity protections and support aspects of Operational Technology (OT) in LNG plants with external vendors providing OT support due to their expertise and knowledge of the vaporization and liquefaction control systems in NG LNG plants. This project will be funded under the KEDNY LNG portfolio while managed by NG IT.

This project has two phases:
1.) Enhancements/Remediation - In FY23, third-party vendor, Network + Security Technologies (NST), is assessing each US LNG plant to identify any vulnerabilities and recommend mitigations. Based on NST's assessments in FY23, this phase will implement operational policy improvements, harden OT networks, and complete other work. All US LNG plants will be addressed in this phase, with TSA-critical plants addressed first.
2.) Support - NG IT will provide ongoing support to all US LNG plants to ensure the enhancements implemented during phase one remain intact. NG IT support will start supporting a given plant after enhancements have been implemented in that plant.

Justification Summary:

Necessary cybersecurity enhancements have been identified in LNG plant operational procedures and OT that will be implemented by this project. This project will also enable LNG plants to comply with the new US federal regulations under Transportation Security Administration's (TSA) Security Directive Pipeline-2021-02C for Pipeline Cybersecurity Mitigation Actions, Contingency Planning, and Testing.

Customer Benefits:

The primary customer benefit is the continuous, safe, and reliable supply of natural gas without unplanned outages due to facility shutdowns. Serving as the principal source of "Peak Shaving" for New York City customers, the plant also plays a strategic role in the company's gas supply portfolio. The facility has two (2) LNG storage tanks with a total design capacity of 1.6 BCF of LNG. Greenpoint LNG facility has a design output of 290 MMSCFD of natural gas. LNG peak-shaving facilities operate intermittently, supplying natural gas pipelines during periods of peak need as required.

Securing OT in all US LNG plants will prevent restoration costs from a cyber-attack that renders a single LNG plant inoperable during the winter heating season, reducing customer bill impacts.

Funding Detail

nationalgrid

Cost Breakdown:

$000	FY25	FY26	FY27	FY28
CAPEX	$ 1,202,398	$	$	$
OPEX				
TOTAL	$ 1,202,398	$	$	$

Supplemental Information

Alternatives

Alternative 1: Do nothing – status quo. This alternative is not recommended because it does not mitigate the potential cybersecurity risks in each US LNG plant.

Alternative 2: Hire a third-party vendor to implement cybersecurity enhancements and provide ongoing support without assessing each LNG plant. This alternative has a lower total cost than the selected approach but is not recommended because it is unlikely to comply with TSA Security Directive Pipeline-2021-02C.

Alternative 3: Non-Pipes Alternatives ("NPAs")

NPA Suitability Screening

1. **Is the project construction expected to commence and be completed within 24 months?**
   ☐ **Yes** ☒ **No**
2. **For KEDLI or NMPC, is the cost of the project less than $500k? For KEDNY, is the cost of the project less than $750k?**
   ☐ **Yes** ☒ **No**
3. **Does the pipes investment affect the critical reliability of the local or broader gas system or respond to a regulatory mandate?**
   ☒ **Yes** ☐ **No**

If questions are answered "yes," the project does not qualify for an NPA.

NPA Evaluation

Has this project been reviewed against the Company's NPA Screening and Suitability Criteria?

☒ **Yes** ☐ **No**

Please briefly explain the results: Project is not suitable for an NPA as it affects the critical reliability of the gas system and mitigates integrity concerns. No NPA, or combination of NPAs, would remove the need for this project at this time.

CLCPA/GHG Analysis

1. **Is the project/program located in an area designated as a disadvantaged community (DAC)[1]?** ☒ Yes ☐ No
2. **If so, explain how was the routing determined and how will the project impact the surrounding area?** Project is within National Grid's property.
3. **Will the project contribute to GHG emission reductions on the Company's gas network?** ☐ Yes ☒ No
4. **Please explain the project's GHG impact.** No Material Impact. This project is required for reliability and is not anticipated to result in GHG emissions reductions.

Studies/References That Support the Program: TSA Security Directive Pipeline-2021-02C.

Benefits
Reliability Benefits ☐ Supports growth forecast ☒ Addresses supply/capacity constraint or supply diversity needs ☐ Addresses storm/climate change resiliency **Other Customer Benefits:** ☐ Improves Customer Satisfaction ☐ Reflects efficiency savings of $____ ☐ Reduction in Billing Errors and Re-Bills ☐ Supports implementation of Gas Business Enablement **Safety Benefits:** ☐ Enhances response time to Emergency Gas Leaks ☒ Enhances employee safety ☐ Increases automation (reduces human error) ☒ Enhances Public Safety ☐ Reduces Damages potential *(i.e., in the case of mapping system)* ☐ Addresses specific safety initiative: _____ **Societal Benefits/Externalities:** ☐ Emissions Reduction ☐ Reduces use of Alt. Fuel ☐ Decreases Leak Prone Pipe \Leak Backlog (Type 3 leaks) ☒ Other: Regulatory Compliance

[1] As defined in Environmental Conservation Law Section 75-0101(5), disadvantaged communities are "communities that bear burdens of negative public health effects, environmental pollution, impacts of climate change, and possess certain socioeconomic criteria, or comprise high-concentrations of low- and moderate- income households, as identified pursuant to section 75-0111 of the [Environmental Conservation Law]."

LNG - Tank 2 Foundation Heaters

Type: ☒ **Project** ☐ **Program**		**Category:** ☒ **Capital** ☐ **Both Capital & O&M**
Investment Code: 5220101204		**Region:** ☒ **KEDNY** ☐ **KEDLI**
Planning Portfolio: ☐ **Customer Connections** ☐ **Mandated** ☒ **Reliability** ☐ **Non-Infrastructure** ☐ **Indirect**		
Annual Program: ☐	**Estimated Start Date:** 03/01/2020	**Estimated In-Service Date:** 6/27
Sponsor: Leyble, Dennis		**Business Contact:** Gray, Kevin

Work Description:

The Greenpoint LNG Facility's Tank 2 has a hot wire loop foundation heating system that was installed in the 1970's and is no longer operating at its original design capacity. Tank 2 is an Liquified Natural Gas (LNG) storage tank that operates at cryogenic temperatures (-260 °F). The foundation heating system prevents the ground from freezing due to these cold temperatures. Several attempts have been made to repair the system by pulling the heating elements and reusing the existing conduits to limited success. Many of the conduits were collapsed or frozen preventing the elements from being replaced. As such the system has continued degrading and requires a long-term solution. This project is to upgrade the tank's existing foundation heating system, restoring it to its original capacity and modernizing the controls.

Justification Summary:

A failure to the foundation heating system could leave the storage tank highly susceptible to frost heaving due to the cryogenic temperatures they are being exposed to. The cryogenic temperatures inside the tank can seep through onto the surface and can cause ice to amass on the ground. Frost heaving can also create cracks in the structure and foundation of the cryogenic tanks which might lead to LNG leaking.

Customer Benefits:

The primary customer benefit is the continuous, safe, and reliable supply of natural gas without unplanned outages due to facility shutdowns. Serving as the principal source of "Peak Shaving" for New York City customers, the plant also plays a strategic role in the company's gas supply portfolio. The facility has two (2) LNG storage tanks with a total design capacity of 1.6 BCF of LNG. Greenpoint LNG facility has a design output of 290 MMSCFD of natural gas. LNG peak shaving facilities operates intermittently, supplying natural gas pipelines during periods of peak need as required.

This project will improve the reliability and safety of the Greenpoint LNG Facility by providing a new tank foundation heating system with modern controls while mitigating the risk of foundation damage or LNG leaks due to frost heave.

Funding Detail

Cost Breakdown:

$000	FY25	FY26	FY27	FY28
CAPEX	$ 10,000	$10,000	$9,000	$4,000

nationalgrid

OPEX				
TOTAL	$ 10,000	$10,000	$9,000	$4,000

Supplemental Information

Alternatives

Alternative 1: Bore new heating conduits under tank. Modern boring technology allows for accurate installation of straight conduits beneath existing structures. This option is to install new conduits and a new electric heating system under Tank 2.

Alternative 2: Complete replacement of under tank heating conduit. Replace the existing foundation heating conduits with a modern equivalent. Tank 2 must be taken out of service, emptied, and purged with fresh air to make safe for entry. The tank floor will be removed, and conduits replaced.

Alternative 3: Do nothing / Defer. The condition of the heater has been deteriorating over time. If this issue is not addressed, or the project deferred the performance of the heating system will continue to decline.

Alternative 4: Non-Pipes Alternatives ("NPAs")

NPA Suitability Screening

1. **Is the project construction expected to commence and be completed within 24 months?**
   ☐ Yes ☒ No
2. **For KEDLI or NMPC, is the cost of the project less than $500k? For KEDNY, is the cost of the project less than $750k?**
   ☐ Yes ☒ No
3. **Does the pipes investment affect the critical reliability of the local or broader gas system or respond to a regulatory mandate?**
   ☒ Yes ☐ No

If questions are answered "yes," the project does not qualify for an NPA.

NPA Evaluation

Has this project been reviewed against the Company's NPA Screening and Suitability Criteria?

☒ Yes ☒ No

Please briefly explain the results: Project is not suitable for an NPA as it affects the critical reliability of the gas system and mitigates integrity concerns. No NPA, or combination of NPAs, would remove the need for this project at this time.

CLCPA/GHG Analysis

1. **Is the project/program located in an area designated as a disadvantaged community (DAC)[1]?** ☒ Yes ☐ No
2. **If so, explain how was the routing determined and how will the project impact the surrounding area?** Project is within National Grid's property. Minimal impacts to community.
3. **Will the project contribute to GHG emission reductions on the Company's gas network?** ☐ Yes ☒ No
4. **Please explain the project's GHG impact.** Direct - Quantifiable. This project is required for reliability. The project is expected to result in increased electricity consumption of approximately 189 MWh/year, which would result in incremental electricity-related emissions of 55 MT CO_2e.

Studies/References That Support the Program:

Benefits
Reliability Benefits
☐ Supports growth forecast
☐ Addresses supply/capacity constraint or supply diversity needs
☐ Addresses storm/climate change resiliency
Other Customer Benefits:
☐ Improves Customer Satisfaction
☐ Reflects efficiency savings of $___
☐ Reduction in Billing Errors and Re-Bills
☐ Supports implementation of Gas Business Enablement
Safety Benefits:
☐ Enhances response time to Emergency Gas Leaks
☒ Enhances employee safety
☒ Increases automation (reduces human error)
☒ Enhances Public Safety
☒ Reduces Damages potential *(i.e., in the case of mapping system)*
☐ Addresses specific safety initiative: _____
Societal Benefits/Externalities:
☐ Emissions Reduction
☐ Reduces use of Alt. Fuel
☐ Decreases Leak Prone Pipe \Leak Backlog (Type 3 leaks)
☐ Other: _____

[1] As defined in Environmental Conservation Law Section 75-0101(5), disadvantaged communities are "communities that bear burdens of negative public health effects, environmental pollution, impacts of climate change, and possess certain socioeconomic criteria, or comprise high-concentrations of low- and moderate- income households, as identified pursuant to section 75-0111 of the [Environmental Conservation Law]."

LNG – Special Projects

Type: ☒ **Project** ☐ **Program**	**Category:** ☒ **Capital** ☐ **Both Capital & O&M**	
Investment Code: 5220001667, 5220101229, 5220001663, 5220001132	**Region:** ☒ **KEDNY** ☐ **KEDLI**	
Planning Portfolio: ☐ **Customer Connections** ☐ **Mandated** ☒ **Reliability** ☐ **Non-Infrastructure** ☐ **Indirect**		
Annual Program: ☐	**Estimated Start Date:** LNG – Nitrogen System Refurbishment: 4/1/2022 LNG – Pump Upgrade (Tank 1): 8/24/2020 LNG – Sub M-Sub L Interconnect: 4/1/2021 LNG – Tail Gas Compressor: 12/4/2017	**Estimated In-Service Date:** LNG – Nitrogen System Refurbishment: 3/31/25 LNG – Pump Upgrade (Tank 1): 3/31/26 LNG – Sub M-Sub L Interconnect: 3/31/25 LNG – Tail Gas Compressor: 6/15/26
Sponsor: Leyble, Dennis	**Business Contact:** Gray, Kevin	

Work Description:

KEDNY has identified several special projects to be completed to ensure the continued safe and reliable operation of the Greenpoint LNG plant. The list of Special Projects includes the following:

- LNG – Nitrogen System Refurbishment [5220001667]
- LNG – Pump Upgrade (Tank 1) [5220101229]
- LNG – Sub M-Sub L Interconnect [5220001663]
- LNG – Tail Gas Compressor [5220001132]

Justification Summary:

LNG – Nitrogen System Refurbishment

This project proposes a replacement of the existing nitrogen system. Over time, the nitrogen tanks and associated equipment are starting to deteriorate, and leaks are becoming more frequent. The tanks are also prone to flooding and outside of the LNG main gate putting them at risk to third party damage. The nitrogen system was identified as a critical component during the Greenpoint Flood Study. As a result, this project will storm harden by relocating the Nitrogen System to a higher elevation.

LNG – Pump Upgrade (Tank 1)

This project aims to upgrade the existing LNG Pumps to improve reliability. There are three low pressure LNG pumps located at Tank 1, each rated to supply 50MMSCFD of LNG to the vaporizers. The pumps are original to the plant and has over 50 years of service. An internal pump inspection and replacement will address issues related to these aging pumps and extend service life of the pumps.

LNG – Sub M-Sub L Interconnect
This project is to bring additional electrical reliability to the LNG facility. Connecting Sub-M to Sub-L will allow an alternate electrical feed source to the facility. Further, this project is considered a storm hardening and reliability effort since both Sub-M and Sub-L is located at higher elevation.

LNG – Tail Gas Compressor
The tail gas compressor has seen a significant increase in maintenance costs and a significant decrease in reliability. The project is to replace the aging tail gas compressor with a new compressor to handle simultaneous flow from LP Flash Gas and Boiloff Gas. Furthermore, this project will storm harden the Jet Compressors and provide a more reliable piece of boil-off handling equipment in the event the Jet Compressors flood or are taken out of service.

Customer Benefits:
The primary customer benefit is the continuous, safe, and reliable supply of natural gas without unplanned outages due to facility shutdowns. Serving as the principal source of "Peak Shaving" for New York City customers, the plant also plays a strategic role in the company's gas supply portfolio. The facility has two (2) LNG storage tanks with a total design capacity of 1.6 BCF of LNG. Greenpoint LNG facility has a design output of 290 MMSCFD of natural gas. LNG peak shaving facilities operates intermittently, supplying natural gas pipelines during periods of peak need as required.

The Nitrogen System Refurbishment will improve plant reliability and safety by replacing obsolete equipment that is prone to leaks and will provide additional automation to the N2 system.

The Pump Upgrade will improve on the plants reliability by replacing aging equipment that is nearing its end of life

The Sub M / Sub L interconnect will address a storm hardening concern by providing an alternative power feed from a source at a higher elevation while also increasing feeder redundancy.

The Tail Gas Compressor will improve on the plant safety, reliability, and contribute to the storm hardening effort by replacing obsolete equipment, increasing plant automation, and adding resiliency to the boiloff system in case the plant's jet compressors are flooded.

Funding Detail

Cost Breakdown:

$000	FY25	FY26	FY27	FY28
CAPEX				
LNG – Nitrogen System Refurbishment	$3,000			
LNG – Pump Upgrade (Tank 1)	$3,590	$100		
LNG – Sub M-Sub L Interconnect	$1,100			
LNG – Tail Gas Compressor	$1,319	$8,000	$10,000	
OPEX				
TOTAL	**$9,009**	**$8,100**	**$10,000**	

Supplemental Information

Alternatives

LNG – Nitrogen System Refurbishment

Alternative 1: Replace Outdated N2 Supply System and Install New System South of LNG Tanks

Replace the existing N2 supply system with new equipment, new instruments and new piping up to tie-in points to the LNG plant systems.

Alternative 2: Do Nothing / Defer Project (Not Recommended)

The N2 supply system is from the 1960s/1970s. There are a record of repairs and leaks on the system. As such it is not recommended to do nothing or defer this project.

LNG – Pump Upgrade (Tank 1)

Alternative 1: Replacing the existing (4-stages) T1 LNG LP pumps with new (5-stages) pumps

This option includes the decommissioning of the existing 4-stages T1 LP pumps and replacing them with new 5-stages pumps with new motors.

Alternative 2: Do Nothing / Defer Project (Not Recommended)

The Tank 1 LNG pumps has been in service since it was commissioned in 1968 with very little upgrades to the system. An overhaul of the pump is necessary to ensure the continuation of reliable service and to achieve future demand targets to meeting customers needs.

LNG – Sub M-Sub L Interconnect

Alternative 1: Connect Sub-M to Sub-L. Maintain existing connectivity from Sub-W and Sub-A.

An electrical feed shall be run from Sub-M to Sub-L. An automatic transfer switch (ATS) shall be provided to swap between the new feed from Sub-M and the existing feed from Sub-W. These feeds from Sub-M and Sub-W shall provide power to Sub-L's East Transformer. The feed from Sub-A shall continue to supply power to Sub-L's West Transformer.

Alternative 2: Do Nothing / Defer Project (Not Recommended)

Doing nothing or deferring this project will leave major company assets in jeopardy. This could potentially disrupt the natural gas pipelines during periods of peak demand.

LNG – Tail Gas Compressor

Alternative 1: Replace the tail gas compressor with a modern equivalent and upgrading its controls.

Alternative 2: Do nothing/defer project (not recommended).

Doing nothing or deferring this project will leave major company assets in jeopardy. This could potentially disrupt the natural gas pipelines during periods of peak demand.

Alternative: Non-Pipes Alternatives ("NPAs")

NPA Suitability Screening

1. **Is the project construction expected to commence and be completed within 24 months?**

   ☐ **Yes** ☒ **No**

2. **For KEDLI or NMPC, is the cost of the project less than $500k? For KEDNY, is the cost of the project less than $750k?**

   ☐ Yes ☒ No

3. **Does the pipes investment affect the critical reliability of the local or broader gas system or respond to a regulatory mandate?**

   ☒ Yes ☐ No

If questions are answered "yes," the project does not qualify for an NPA.

NPA Evaluation

Has this project been reviewed against the Company's NPA Screening and Suitability Criteria?

☒ Yes ☐ No

Please briefly explain the results: Project is not suitable for an NPA as it affects the critical reliability of the gas system and mitigates integrity concerns. No NPA, or combination of NPAs, would remove the need for this project at this time.

CLCPA/GHG Analysis

1. **Is the project/program located in an area designated as a disadvantaged community (DAC)[1]?** ☒ Yes ☐ No
2. **If so, explain how was the routing determined and how will the project impact the surrounding area?** Project is within National Grid's property. Minimal impacts to community.
3. **Will the project contribute to GHG emission reductions on the Company's gas network?** ☐ Yes ☒ No
4. **Please explain the project's GHG impact.** Direct - Quantifiable. This project is required for reliability. The Tail Gas Compressor upgrade is expected to result in increased electricity consumption of approximately 20 MWh/year, which would result in incremental electricity-related emissions of 6 MT CO_2e.

Studies/References That Support the Program:

[1] As defined in Environmental Conservation Law Section 75-0101(5), disadvantaged communities are "communities that bear burdens of negative public health effects, environmental pollution, impacts of climate change, and possess certain socioeconomic criteria, or comprise high-concentrations of low- and moderate- income households, as identified pursuant to section 75-0111 of the [Environmental Conservation Law]."

Benefits
Reliability Benefits ☒ Supports growth forecast ☐ Addresses supply/capacity constraint or supply diversity needs ☐ Addresses storm/climate change resiliency **Other Customer Benefits:** ☐ Improves Customer Satisfaction ☐ Reflects efficiency savings of $___ ☐ Reduction in Billing Errors and Re-Bills ☐ Supports implementation of Gas Business Enablement **Safety Benefits:** ☐ Enhances response time to Emergency Gas Leaks ☒ Enhances employee safety ☒ Increases automation (reduces human error) ☐ Enhances Public Safety ☐ Reduces Damages potential *(i.e., in the case of mapping system)* ☐ Addresses specific safety initiative: _____ **Societal Benefits/Externalities:** ☐ Emissions Reduction ☐ Reduces use of Alt. Fuel ☐ Decreases Leak Prone Pipe \Leak Backlog (Type 3 leaks) ☐ Other: _____

Pipeline Integrity - IMP - Jamaica Bay Line ILI

Type: ☒ **Project** ☐ **Program**	**Category:** ☒ **Capital** ☐ **Both Capital & O&M**	
Investment Code: 5220001843	**Region:** ☒ **KEDNY** ☐ **KEDLI**	
Planning Portfolio: ☐ **Customer Connections** ☒ **Mandated** ☐ **Reliability** ☐ **Non-Infrastructure** ☐ **Indirect**		
Annual Program: ☐	**Estimated Start Date: In - Progress**	**Estimated In-Service Date: 3/26**
Sponsor: Kern, Mike	**Business Contact:** Conklin, Kevin	

Work Description:

Jamaica Bay Line - 0.1 - Robotic ILI Enablement --- Work scope includes installation of large diameter Hot Tap fittings, replace existing plug valves, install charging ports and permanent vaults to facilitate inspection on a seven-year cycle.

Justification Summary:

Pursuant to the 2002 Act, the DOT promulgated rules on managing the integrity of transmission pipelines used by the gas and hazardous liquids industries under 49 CFR Part 192.901 – 192.951, which became effective on January 14, 2004. These regulations require pipeline operators to develop and implement an IMP for "covered" transmission pipelines, which are defined as certain pipelines in high consequence areas (HCA). The program required that the first cycle of pipeline assessments be completed no later than 2012. Reassessments are required to be completed at intervals not exceeding seven years thereafter from the last assessment. The assessments are comprised of external corrosion direct assessment (ECDA) and ILI. The results of each operator's program are summarized and reported to the DOT on an annual basis.

Pipeline safety laws and regulations constantly evolve driving progressive changes in utility operations and asset management. San Bruno and several other high profile pipeline incidents have set in motion recommendations, proposed rulemaking, and the 2011 Act signed into law on January 3, 2012. In 2019 &2022 PHMSA issued two rules (RIN 1 & RIN 2) that will address the 2011 Act mandates and implement several additional changes to the regulations for gas pipelines. The rule includes the following significant items that will affect the IMP & IVP programs:

- Advised operators to make all pipeline segments operating at or over 20 percent SMYS ILI enabled

Customer Benefits:

ILI enablement will allow for the enhanced operational safety of the Company's transmission system. Through ILI enablement, the Company can proactively inspect existing transmission assets to identify anomalies or other issues that might adversely affect the integrity of the system and result in gas safety concerns for the system and our customers while meeting regulatory requirements.

Funding Detail

Cost Breakdown:

ILI enablement will allow for the enhanced operational safety of the Company's transmission system. Through ILI enablement, the Company can proactively inspect existing transmission assets to identify anomalies or other issues that might adversely affect the integrity of the system and result in gas safety concerns for the system and our customers.

$000	FY25	FY26	FY27	FY28
CAPEX	$ 18,966	$ 15,000	$ -	$ -
OPEX				
TOTAL	$ 18,966	$ 15,000	$ -	$ -

Supplemental Information

Alternatives

Alternative 1: Maintain Current IMP using ECDA testing method. This does not support PHMSA's (NTSB) recommendation that operators retrofit pipelines to be ILI enabled.

Alternative 2: Non-Pipes Alternatives ("NPAs")

NPA Suitability Screening

1. **Is the project construction expected to commence and be completed within 24 months?**
   ☐ **Yes** ☒ **No**
2. **For KEDLI or NMPC, is the cost of the project less than $500k? For KEDNY, is the cost of the project less than $750k?**
   ☐ **Yes** ☒ **No**
3. **Does the pipes investment affect the critical reliability of the local or broader gas system or respond to a regulatory mandate?**
   ☒ **Yes** ☐ **No**

If questions are answered "yes," the project does not qualify for an NPA.

NPA Evaluation

Has this project been reviewed against the Company's NPA Screening and Suitability Criteria?
☒ **Yes** ☐ **No**

Please briefly explain the results: This mandated work is required by federal regulations and cannot be replaced with an NPA.

CLCPA/GHG Analysis

1. **Is the project/program located in an area designated as a disadvantaged community (DAC)[1]?** ☒ Yes ☐ No
2. **If so, explain how was the routing determined and how will the project impact the surrounding area?** Project location was selected based on location of existing assets and could not be avoided. The project will only have a

[1] As defined in Environmental Conservation Law Section 75-0101(5), disadvantaged communities are "communities that bear burdens of negative public health effects, environmental pollution, impacts of climate change, and possess certain socioeconomic criteria, or comprise high-concentrations of low- and moderate- income households, as identified pursuant to section 75-0111 of the [Environmental Conservation Law]."

temporary impact on the surrounding area and, where possible, the Company will seek to minimize its time at the location.

3. **Will the project contribute to GHG emission reductions on the Company's gas network?** ☒ Yes ☐ No
4. **Please explain the project's GHG impact.** Direct - Non-Quantifiable. The project will proactively review assets and has the potential to avoid future leaks and failures on the system. Therefore, it could have a nominal impact on GHG reductions if a leak was discovered or avoided early because of the project.

Studies/References That Support the Program:

Benefits
Reliability Benefits
☐ Supports growth forecast
☐ Addresses supply/capacity constraint or supply diversity needs
☐ Addresses storm/climate change resiliency
Other Customer Benefits:
☐ Improves Customer Satisfaction
☐ Reflects efficiency savings of $___
☐ Reduction in Billing Errors and Re-Bills
☐ Supports implementation of Gas Business Enablement
Safety Benefits:
☐ Enhances response time to Emergency Gas Leaks
☐ Enhances employee safety
☐ Increases automation (reduces human error)
☒ Enhances Public Safety
☐ Reduces Damages potential *(i.e., in the case of mapping system)*
☐ Addresses specific safety initiative: _____
Societal Benefits/Externalities:
☒ Emissions Reduction
☐ Reduces use of Alt. Fuel
☐ Decreases Leak Prone Pipe \Leak Backlog (Type 3 leaks)
☒ Other: Regulatory compliance

Exhibit ___ (GIOP-7CU):
Revised Incremental O&M Non-Labor Expenditures for the Rate Year and Data Years

The Brooklyn Union Gas Company d/b/a National Grid NY
Incremental Operating Expenses
Gas Safety and Gas Infrastructure and Operations Panels
$000's

Program Description	Rate Year Ending March 31, 2025			Data Year Ending March 31, 2026			Data Year Ending March 31, 2027			Data Year Ending March 31, 2028		
	Non-labor	Labor	Total	Non-labor	Labor	Total	Non-labor	Labor	Total	Non-labor	Labor	Total
Gas Safety Programs												
Advanced Communications Infrastructure (AMI)	$ 1,000.0	$ -	$ 1,000.0	$ 1,000.0	$ -	$ 1,000.0	$ 434.0	$ 16.9	$ 450.9	$ 670.0	$ 17.4	$ 687.4
Advanced Communications Infrastructure (RMD)	759.2	17.0	776.2	1,485.6	17.6	1,503.2	2,087.8	18.1	2,105.9	2,390.2	37.2	2,427.4
Advanced Communications Infrastructure (Smart Meter Repl.)	-	-	-	-	-	-	8.6	48.7	57.3	25.8	87.4	113.2
Advanced Leak Detection	2,761.6	489.2	3,250.8	2,821.8	499.4	3,321.2	2,880.8	509.6	3,390.5	2,940.2	519.8	3,460.0
Inside Service Line Inspections (Bus Dist)	-	-	-	(300.0)	-	(300.0)	(600.0)	-	(600.0)	(995.0)	-	(995.0)
Inside Service Line Inspections (Non-bus Dist)	1,321.0	-	1,321.0	(4,378.6)	-	(4,378.6)	(2,371.1)	-	(2,371.1)	(7,586.0)	-	(7,586.0)
Damage Prevention Program Enhancements	1,146.4	-	1,146.4	1,533.3	-	1,533.3	1,932.1	-	1,932.1	2,343.1	-	2,343.1
Operator Qualification Program	1,335.1	1,319.3	2,654.3	1,346.9	1,356.1	2,703.0	1,371.5	1,393.3	2,764.8	1,395.9	1,430.8	2,826.7
Gas Safety Public Awareness	288.2	-	288.2	293.7	-	293.7	299.1	-	299.1	304.4	-	304.4
I&R – Low Pressure Valve Inspection	-	153.6	153.6	-	327.7	327.7	-	588.1	588.1	-	622.4	622.4
Gas Pipeline Safety & Compliance	-	121.1	121.1	-	124.8	124.8	-	128.6	128.6	-	132.4	132.4
Gas Pipeline Safety and Control Testing	-	132.6	132.6	-	136.7	136.7	-	140.8	140.8	-	145.0	145.0
Gas Pipeline Safety Management	-	121.3	121.3	-	125.0	125.0	-	128.8	128.8	-	132.6	132.6
Gas Pipeline Safety Q&A	-	162.1	162.1	-	167.1	167.1	-	172.1	172.1	-	177.2	177.2
Meter Relocation Program	-	19.8	19.8	-	20.2	20.2	-	28.0	28.0	-	28.6	28.6
Voluntary Integrity Management Program	1,000.0	276.0	1,276.0	1,019.4	281.8	1,301.1	1,038.0	287.5	1,325.5	1,056.5	293.2	1,349.7
Sub-total	$ 9,611.4	$ 2,811.9	$ 12,423.3	$ 4,822.0	$ 3,056.4	$ 7,878.5	$ 7,080.8	$ 3,460.6	$ 10,541.4	$ 2,545.1	$ 3,624.1	$ 6,169.2
GIOP Programs												
Pipeline Integrity - IMP/ IVP	$ 601.0	$ -	$ 601.0	$ (26.6)	$ -	$ (26.6)	$ (2,249.7)	$ -	$ (2,249.7)	$ (1,394.6)	$ -	$ (1,394.6)
Disconnects & Reconnects	2,413.7	-	2,413.7	3,240.5	-	3,240.5	3,772.8	-	3,772.8	4,227.4	-	4,227.4
Tools & Equipment	550.0	-	550.0	590.0	-	590.0	685.0	-	685.0	780.0	-	780.0
Storm Hardening	-	-	-	-	-	-	-	-	-	899.5	-	899.5
City State Construction	-	50.3	50.3	-	77.7	77.7	-	90.6	90.6	-	97.6	97.6
Gas Control	-	212.5	212.5	-	304.3	304.3	-	313.5	313.5	-	322.8	322.8
Pressure Reg/LNG/CNG	-	13.3	13.3	-	13.7	13.7	-	14.2	14.2	-	14.6	14.6
Resource Coordination	-	102.3	102.3	-	126.6	126.6	-	130.4	130.4	-	134.3	134.3
Stakeholder Engagement	-	12.9	12.9	-	26.6	26.6	-	27.4	27.4	-	28.2	28.2
Storm Hardening	-	-	-	-	-	-	-	-	-	-	52.8	52.8
Transmission Engineering	-	17.3	17.3	-	17.8	17.8	-	18.3	18.3	-	18.9	18.9
Investment Planning	-	3.8	3.8	-	4.7	4.7	-	4.9	4.9	-	5.0	5.0
ERT Replacement Program	-	51.3	51.3	-	62.1	62.1	-	73.2	73.2	-	84.7	84.7
Collections - CMS	-	1,865.1	1,865.1	-	1,904.1	1,904.1	-	1,943.0	1,943.0	-	1,981.6	1,981.6
Sub-total	$ 3,564.7	$ 2,328.7	$ 5,893.5	$ 3,803.9	$ 2,537.6	$ 6,341.5	$ 2,208.0	$ 2,615.5	$ 4,823.5	$ 4,512.3	$ 2,740.5	$ 7,252.9

The Brooklyn Union Gas Company d/b/a National Grid NY
Incremental Operating Expenses
Gas Safety and Gas Infrastructure and Operations Panels
$000's

Program Description	Rate Year Ending March 31, 2025			Data Year Ending March 31, 2026			Data Year Ending March 31, 2027			Data Year Ending March 31, 2028		
	Non-labor	Labor	Total	Non-labor	Labor	Total	Non-labor	Labor	Total	Non-labor	Labor	Total
GIOP Future of Heat Programs												
Future of Heat - Hydrogen Blending	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ 250.0	$ -	$ 250.0
Future of Heat - RNG	-	-	-	-	-	-	-	-	-	500.0	-	500.0
Hydrogen	-	53.0	53.0	-	54.7	54.7	-	56.3	56.3	-	58.0	58.0
Materials Lab	-	134.6	134.6	-	138.2	138.2	-	141.9	141.9	-	145.6	145.6
Research & Development	-	292.5	292.5	-	301.5	301.5	-	310.6	310.6	-	319.8	319.8
Gas Quality Lab	-	85.5	85.5	-	87.7	87.7	-	89.8	89.8	-	92.0	92.0
RNG	-	31.8	31.8	-	32.8	32.8	-	33.8	33.8	-	34.8	34.8
Geothermal	-	145.1	145.1	-	149.5	149.5	-	293.4	293.4	-	301.1	301.1
Newtown Creek Project	1,062.6	-	1,062.6	1,090.5	-	1,090.5	1,119.1	-	1,119.1	1,148.3	-	1,148.3
Sub-total	$ 1,062.6	$ 742.5	$ 1,805.1	$ 1,090.5	$ 764.4	$ 1,854.9	$ 1,119.1	$ 925.9	$ 2,045.0	$ 1,898.3	$ 951.4	$ 2,849.7
GIOP Fleet & Facilities												
Fleet Asset & Performance	$ 504.7	$ 12.4	$ 517.1	$ 552.8	$ 12.8	$ 565.6	$ 695.2	$ 13.2	$ 708.4	$ 682.3	$ 13.6	$ 695.9
Warehouse DNY	-	95.3	95.3	-	97.2	97.2	-	99.2	99.2	-	101.2	101.2
Facilities DNY	-	101.4	101.4	-	103.5	103.5	-	105.6	105.6	-	107.7	107.7
Sub-total	$ 504.7	$ 209.1	$ 713.8	$ 552.8	$ 213.6	$ 766.4	$ 695.2	$ 218.1	$ 913.2	$ 682.3	$ 222.5	$ 904.8
Grand Total	$ 14,743.4	$ 6,092.2	$ 20,835.7	$ 10,269.2	$ 6,572.0	$ 16,841.2	$ 11,103.1	$ 7,220.0	$ 18,323.2	$ 9,638.1	$ 7,538.5	$ 17,176.6

Exhibit ___ (GIOP-8CU):
Revised Incremental FTE Positions by Function in the Rate Year and Data Years

The Brooklyn Union Gas Company d/b/a National Grid NY
Incremental FTE's
Gas Safety and Gas Infrastructure and Operations Panels

	Rate Year Ending March 31, 2025	Data Year Ending March 31, 2026	Data Year Ending March 31, 2027	Data Year Ending March 31, 2028
Gas Safety Programs				
Advanced Communications Infrastructure (AMI)	-	-	2.0	2.0
Advanced Communications Infrastructure (RMD)	1.0	1.0	1.0	2.0
Advanced Communications Infrastructure (Smart Meter Repl.)	-	-	8.0	15.0
Advanced Leak Detection	4.0	4.0	4.0	4.0
Operator Qualification Program	16.0	16.0	16.0	16.0
I&R – Low Pressure Valve Inspection	2.0	4.0	7.0	7.0
Gas Pipeline Safety & Compliance	1.0	1.0	1.0	1.0
Gas Pipeline Safety and Control Testing	2.0	2.0	2.0	2.0
Gas Pipeline Safety Management	1.5	1.5	1.5	1.5
Gas Pipeline Safety Q&A	2.0	2.0	2.0	2.0
Meter Relocation Program	3.0	3.0	4.0	4.0
Voluntary Integrity Management Program	2.0	2.0	2.0	2.0
Sub-total	34.5	36.5	50.5	58.5
GIOP Programs				
City State Construction	10.0	12.0	13.0	13.0
Gas Control	2.5	2.5	2.5	2.5
Pressure Reg/LNG/CNG	1.5	1.5	1.5	1.5
Resource Coordination	3.0	3.0	3.0	3.0
Stakeholder Engagement	1.0	2.0	2.0	2.0
Storm Hardening	-	-	-	8.0
Transmission Engineering	1.0	1.0	1.0	1.0
Investment Planning	0.5	0.5	0.5	0.5
ERT Replacement Program	14.0	17.0	20.0	23.0
Collections - CMS	30.0	30.0	30.0	30.0
Sub-total	63.5	69.5	73.5	84.5
GIOP Future of Heat Programs				
Hydrogen	0.5	0.5	0.5	0.5
Materials Lab	1.5	1.5	1.5	1.5
Research & Development	1.5	1.5	1.5	1.5
Gas Quality Lab	1.0	1.0	1.0	1.0
RNG	1.5	1.5	1.5	1.5
Geothermal	4.0	4.0	5.0	5.0
Sub-total	10.0	10.0	11.0	11.0
GIOP Fleet & Facilities				
Fleet Asset & Performance	0.3	0.3	0.3	0.3
Warehouse DNY	2.0	2.0	2.0	2.0
Facilities DNY	2.0	2.0	2.0	2.0
Sub-total	4.3	4.3	4.3	4.3
Grand Total	112.3	120.3	139.3	158.3

Exhibit ___ (GIOP-10CU):
Revised Facilities Service Company Capital Forecast

The Brooklyn Union Gas Company d/b/a National Grid NY
GIOP-10CU - Facilities Service Company Capital Forecast

Program	*FY23 CAPEX	FY24 CAPEX	FY25 CAPEX	FY26 CAPEX	FY27 CAPEX	FY28 CAPEX
Asset Renewal - Replacement	$ 6,879,766	$ 3,272,000	$ 16,588,999	$ 1,933,500	$ 2,403,500	$ 4,003,500
Base	$ 11,542,532	$ 13,115,401	$ 1,000,000	$ 1,000,000	$ 1,000,000	$ 1,000,000
EV Charging	$ -	$ -	$ 1,500,000	$ 2,755,000	$ -	$ -
**LI Warehouse	$ -	$ -	$ 4,000,000	$ -	$ -	$ -
**RE Strategy	$ 1,009,953	$ 500,000	$ 5,000,000	$ -	$ -	$ -
Safety	$ 3,870,010	$ 14,112,058	$ 5,247,230	$ 2,172,653	$ 1,225,000	$ 1,150,000
Grand Total	$ 23,302,261	$ 30,999,459	$ 33,336,229	$ 9,696,153	$ 4,628,500	$ 6,153,500

*FY23 = CWIP balance through December 31 + 3 months Forecast

**LI Warehouse and RE- Strategy (FY24-25) investments impact KEDLI primarily

***Revenue Requirement impacts of these programs capital investments can be found in Exhibit ___ (RRP-11CU), Workpapers to RRP-3CU, Schedule 9, Workpapers 7, 10 and 13

Exhibit ___ (GIOP-11CU):
Revised Fleet Electrification Costs

The Brooklyn Union Gas Company d/b/a National Grid NY
Fleet Electrification Costs
($000's)

Type Of Spend	Rate Year Ending March 31, 2025	Data Year Ending March 31, 2026	Data Year Ending March 31, 2027	Data Year Ending March 31, 2028	Source Exhibit/Workpaper
Direct Capital Spend					
Fixed EV Chargers	$ 4,245.0	$ 1,135.0	$ 2,795.0	$ 3,440.0	GIOP-1CU
Total Direct Capital Spend	$ 4,245.0	$ 1,135.0	$ 2,795.0	$ 3,440.0	
Incremental Operating Expense					
EV Incremental Lease Expense	$ 61.2	$ 138.3	$ 186.6	$ 220.4	RRP-3CU, Schedule 24
Battery EV Charger Lease Expense	$ 53.0	$ 123.3	$ 110.9	$ 114.5	RRP-3CU, Schedule 24
Electricity Consumption	$ 22.9	$ 79.7	$ 88.7	$ 133.4	RRP-3CU, Schedule 40
Electric Infrastructure Make Ready	$ 400.0	$ 105.9	$ -	$ -	RRP-3CU, Schedule 40
Service Company Rent Expense - Fixed EV Chargers [1]	$ -	$ 5.1	$ 24.9	$ 23.1	RRP-3CU, Schedule 9
Service Company Rent Expense - Site Backup Generation [2]	$ -	$ 213.1	$ 408.6	$ 378.3	RRP-3CU, Schedule 9
Sub-total	$ 537.1	$ 665.4	$ 819.7	$ 869.8	
Reduced Operating Expense					
Reduced Fuel Consumption	$ (32.4)	$ (112.5)	$ (124.6)	$ (187.5)	RRP-3CU, Schedule 24
Sub-total	$ (32.4)	$ (112.5)	$ (124.6)	$ (187.5)	
Total Operating Expense	$ 504.7	$ 552.8	$ 695.2	$ 682.3	

Notes:
1 - Rent expense resulting from ServCo investment in Fixed EV Chargers of $1.255MM in FY27
2 - Rent expense resulting from ServCo investment in mobile charging solution of $1.5MM in FY25 and $1.5MM in FY26

Appendix A
Company Responses to Information Requests That Discuss Corrections and Updates

Date of Request: May 17, 2023 Request No. DPS-273
Due Date: May 30, 2023 NG Request No. NG-276

KeySpan Gas East Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY

KEDNY KEDLI Case 23-G-0225/0226 Data Requests

Request for Information

FROM: John Castano

TO: National Grid

SUBJECT: Other Initiatives

Request:

In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

KEDNY anticipates hiring 133.9 full-time equivalent (FTE) employees, and KEDLI anticipates hiring 99.6 FTE employees by the end of the rate year.

1. For each FTE who had an anticipated start date prior to May 17, 2023, as shown on Exhibit__(RRP-11), Workpapers to Exhibit RRP-3, Schedule 41, Workpaper 2, provide the actual start date and the number of days between the anticipated start date and the actual start date.

    a. Identify whether each FTE position with an anticipated start date prior to May 17, 2023, is currently filled.

    b. Identify whether each FTE position with an anticipated start date prior to May 17, 2023, was filled internally or externally.

2. For each FTE position with an anticipated start date prior to May 17, 2023, that is not currently filled, explain why the position is not filled.

3. For any FTE positions that have been filled internally, has the Company back-filled the vacated position?

4. Provide monthly updates to the Company's responses to the questions above.

Response:

1. See Attachment 1, column “Z” for the FTE actual hire date and column “AE” for count of days between the anticipated hire date and actual hire date.

a. See Attachment 1, column "Y". Also see column "AK" for additional notes that indicate the Company will be correcting the start dates of 73 positions and column "G" for the corrected Start Date. Because all dates impacted by the correction are prior to the start of the Rate Year, the correction is presentation only and has no revenue requirement impact. The correction will be made to Exhibit__(RRP-11), Workpapers to Exhibit RRP-3, Schedule 41, Workpaper 2 in Corrections and Updates.

b. See Attachment 1, column "AA" for an indication of whether a FTE was filled internally or externally.

2. See Attachment 1, column "AC."

3. See Attachment 1, column "AB."

4. The Company will continue to respond to this information request on a monthly basis.

Name of Respondent:	Date of Reply:
Mark Stiner	May 30, 2023

Date of Request: May 30, 2023 Request No. DPS-360
Due Date: June 9, 2023 NG Request No. NG-370

KeySpan Gas East Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY

KEDNY KEDLI Case 23-G-0225/0226 Data Requests

Request for Information

FROM: Robert Cully

TO: National Grid

SUBJECT: Light Duty EV Fleet Deployment Schedule

Request:

In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

On p. 92 of the KEDNY Gas Infrastructure and Operations Panel (GIOP) initial testimony, KEDNY states that it proposes to replace 413 KEDNY-leased internal combustion engine (ICE) vehicles and 36 leased Service Company vehicles with an equivalent number of electric vehicles (EVs) during the Rate Year and Data Years. On p. 92 of the KEDLI GIOP initial testimony, KEDLI states that it proposes to replace 234 KEDLI-leased ICE vehicles and 36 leased Service Company vehicles with an equivalent number of EVs during the Rate Year and Data Years.

1. Explain whether the 36 Service Company-leased vehicles referred to by KEDNY and KEDLI are the same vehicles, i.e. are there 36 total Service Company vehicles the Companies plan to replace, 72 Service Company vehicles, or some other number?

2. For KEDNY, provide the number of ICE vehicles the Company proposes to replace with EVs during each of the following periods:

    a. The Rate Year;
    b. Data Year 1;
    c. Data Year 2; and
    d. Data Year 3.

3. For KEDLI, provide the number of ICE vehicles that the Company proposes to replace during each of the following periods:

    a. The Rate Year;
    b. Data Year 1;
    c. Data Year 2; and
    d. Data Year 3.

4. Regarding the Service Company, provide the number of ICE vehicles that the Company proposes to replace during each of the following periods:

   a. The Rate Year;
   b. Data Year 1;
   c. Data Year 2; and
   d. Data Year 3.

Response:

1. Following the Companies' rate filing in April, National Grid reviewed its overall fleet electrification plans as part of its long-term fleet electrification strategy. A lack of available fit-for-purpose vehicles and marketplace uncertainty/unavailability drove the decision to reduce the number of electric vehicles ("EVs") and associated chargers being proposed in these rate cases. The Companies will provide these revised figures, with an explanation for the changes, as part of the Corrections and Updates filing. The numbers presented in this response reflect the revised plan.

   The proposed conversion of 36 Service Company-leased vehicles to EVs in both the KEDNY and KEDLI rate filings were referring to the same vehicles. In total, 36 Service Company vehicles were proposed for conversion to EVs, not 72. Based on the updated fleet electrification plan, 18 Service Company vehicles are now being proposed for conversion to EV.

2. The table below reflects the number of internal combustion engine ("ICE") vehicles the Company proposes to replace with EVs for KEDNY as part of its revised fleet electrification plan.

	RY (a)	**DY1 (b)**	**DY2 (c)**	**DY3 (d)**
KEDNY	33	85	11	67

3. The table below reflects the number of ICE vehicles the Company proposes to replace with EVs for KEDLI as part of its revised fleet electrification plan.

	RY (a)	**DY1 (b)**	**DY2 (c)**	**DY3 (d)**
KEDLI	38	24	17	15

4. The table below reflects the number of ICE vehicles the Company proposes to replace with EVs for Service Company-leased vehicles as part of its revised fleet electrification plan.

	RY (a)	**DY1 (b)**	**DY2 (c)**	**DY3 (d)**
SERVCO	6	0	12	0

Name of Respondent:
Brian Schiavone
Mark Stiner

Date of Reply:
June 7, 2023

Date of Request: May 30, 2023 Request No. DPS-362
Due Date: June 9, 2023 NG Request No. NG-372

KeySpan Gas East Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY

KEDNY KEDLI Case 23-G-0225/0226 Data Requests

Request for Information

FROM: Robert Cully

TO: National Grid

SUBJECT: Level 2 and DCFC EV Charger Deployment at Company Sites

Request:

In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

On p. 92 of the KEDNY Gas Infrastructure and Operations Panel (GIOP) initial testimony, KEDNY states that it proposes to install 430 Level 2 electric vehicle (EV) charging ports and 31 Direct Current Fast Charging (DCFC) EV charging ports at five KEDNY locations, as well as Level 2 and DCFC ports at other sites with an allocation to KEDNY. On pp. 92-93 of the KEDLI GIOP initial testimony, KEDLI states that it proposes to install 151 Level 2 EV charging ports and 28 DCFC EV charging ports at "various" KEDLI locations, as well as Level 2 and DCFC ports at other sites with an allocation to KEDLI.

1. Explain how the Companies developed their forecasts of the amount of Level 2 and DCFC chargers needed to support their light duty fleet electrification plans.

2. Provide a list of each KEDNY site at which EV charging ports will be installed, and identify the number of each plug type (Level 2 and DCFC) that will be installed at each site.

3. Provide a list of each KEDLI site at which EV charging ports will be installed, and identify the number of each plug type (Level 2 and DCFC) will be installed at each site.

4. For each site identified in response to Questions 2 and 3, above, explain whether the site is either: (a) within a Disadvantaged Community, (b) adjacent to a Disadvantaged Community, or (c) neither within nor adjacent to a Disadvantaged Community.

5. Provide the number of Level 2 chargers installed at KEDNY sites each year for:

    a. The Rate Year;
    b. Data Year 1;
    c. Data Year 2; and
    d. Data Year 3.

6. Provide the number of Level 2 chargers installed at other sites with an allocation to KEDNY each year for:

    a. The Rate Year;
    b. Data Year 1;
    c. Data Year 2; and
    d. Data Year 3.

7. Provide the number of Level 2 chargers installed at KEDLI sites each year for:

    a. The Rate Year;
    b. Data Year 1;
    c. Data Year 2; and
    d. Data Year 3.

8. Provide the number of Level 2 chargers installed at other sites with an allocation to KEDLI each year for:

    a. The Rate Year;
    b. Data Year 1;
    c. Data Year 2; and
    d. Data Year 3.

9. Provide the number of DCFC chargers installed at KEDNY sites each year for:

    a. The Rate Year;
    b. Data Year 1;
    c. Data Year 2; and
    d. Data Year 3.

10. Provide the number of DCFC chargers installed at other sites with an allocation to KEDNY each year for:

    a. The Rate Year;
    b. Data Year 1;
    c. Data Year 2; and
    d. Data Year 3.

11. Provide the number of DCFC chargers installed at KEDLI sites each year for:

    a. The Rate Year;
    b. Data Year 1;
    c. Data Year 2; and
    d. Data Year 3.

12. Provide the number of DCFC chargers installed at other sites with an allocation to KEDLI each year for:

a. The Rate Year;
b. Data Year 1;
c. Data Year 2; and
d. Data Year 3.

13. Explain whether the Level 2 and DCFC chargers installed at sites with an allocation to KEDNY and/or KEDLI are located within the KEDNY or KEDLI service territories.

    a. For any chargers installed at sites that are not within the KEDNY or KEDLI service territory, explain why the Company decided to allocate costs to KEDNY and KEDLI customers.

Response:

1. The Companies determined how many Level 2 and DCFC chargers were needed to support the proposed fleet EV program by analyzing the anticipated fleet replacement schedule over the Rate Year and Data Years, and determining which of the existing leased vehicles were able to be replaced with an EV available in the marketplace. The Companies determined the number and type of chargers required at each facility based on those findings and developed an annual plan for the installation of chargers prior to EVs being assigned to those locations.

   The proposed number of Level 2 chargers corresponds to light-duty vehicles being proposed for conversion to an EV to allow for dedicated overnight charging. The Companies are proposing one DCFC station for approximately every 10 light-duty vehicles proposed for conversion to an EV to allow for opportunistic charge during normal working hours. The ratios were recommended by an external consultant and are consistent with similarly sized fleet EV conversions.

   Following the Companies' rate filing in April, National Grid reviewed its overall fleet electrification plans as part of its long-term fleet electrification strategy. A lack of available fit-for-purpose EVs and marketplace uncertainty/unavailability drove the decision to reduce the number of EVs and associated chargers being proposed. The Companies will provide the revised figures, with an explanation for the changes, as part of the Corrections and Updates filing. The numbers presented in this response reflect the revised plan.

   Under the revised plan, the Companies propose to install 193 Level 2 and 29 DCFC stations at five KEDNY locations, 106 Level 2 and 14 DCFC stations at 11 KEDLI locations, and 21 Level 2 and three DCFC stations at two Service Company sites (Brentwood and Melville) with an allocation to KEDLI and KEDNY as outlined in Exhibit ___ (GIOP-10), and set forth in more detail in Exhibit ___ (RRP-11), Schedule 9, Workpapers 4, 7, 10, an 13.

2. The number of proposed Level 2 and DCFC stations, by site, for KEDNY are:

KEDNY Location	Level 2 Charging Stations	DCFC Stations
Canarsie	97	12
Greenpoint	60	10
Staten Island	23	5

KEDNY Location	Level 2 Charging Stations	DCFC Stations
Springfield	11	2
Bay Ridge / Brooklyn	2	0

3. The number of proposed Level 2 and DCFC stations, by site, for KEDLI are:

KEDLI Location	Level 2 Charging Stations	DCFC Charging Stations
Bellmore	29	4
Hewlett	13	3
Roslyn	13	2
Patchogue	15	2
Hicksville	9	2
Brightwaters	9	1
Brentwood	8	0
Shoreham	4	0
Glenwood	2	0
Holtsville	2	0
Riverhead	2	0

4. Sites within or adjacent to a Disadvantaged Community included in the response to Question 2 (KEDNY) are as follows:

Canarsie – adjacent to Disadvantaged Communities (Brownsville, East Flatbush, Canarsie)
Greenpoint – within a Disadvantaged Community
Staten Island – within a Disadvantaged Community
Springfield (Jamaica) – adjacent to a Disadvantaged Community (Locust Manor)
Bay Ridge – adjacent to a Disadvantaged Community (other sections of Bay Ridge)

Sites within or adjacent to a Disadvantaged Community included in the response to Question 3 (KEDLI) are as follows:

Bellmore – adjacent to a Disadvantaged Community (Freeport)
Brentwood – adjacent to a Disadvantaged Community (Central Islip and Brentwood)
Brightwaters – adjacent to a Disadvantaged Community (Bay Shore)
Glenwood – adjacent to a Disadvantaged Community (Sea Cliff and Glen Cove)
Hewlett – within and adjacent to a Disadvantaged Community (Valley Stream)
Holtsville – neither within nor adjacent to a Disadvantaged Community
Hicksville – adjacent to a Disadvantaged Community (New Cassel)
Patchogue – within a Disadvantaged Community
Shoreham – neither within nor adjacent to a Disadvantaged Community
Riverhead – within a Disadvantaged Community
Roslyn – neither within nor adjacent to a Disadvantaged Community

5.-12. Please see Attachment 1 for the information requested in questions 5 through 12.

13. and 13. a.

The Service Company will own Level 2 and DCFC stations that will be partially allocated to KEDNY and KEDLI. The Service Company charging stations will be installed within KEDLI's service territory; none are currently planned for KEDNY's service territory.

The costs associated with the Service Company charging stations are appropriately allocated to both KEDNY and KEDLI because these facilities will provide charging capabilities for EVs that support personnel while traveling between the service territories for meetings and other business purposes. For example, KEDNY personnel will be able utilize charging stations when receive training at National Grid sites in KEDLI's service territory.

Name of Respondent: Therese Sullivan

Date of Reply: June 7, 2023

SERVCO

Jurisdiction	Location Name	Total Level 2	Total DCFC	The Rate Year - L2 Charger	The Rate Year - DCFC Ch	Data Year 1 Level 2	Data Year 1 DCFC	Data Year 2 Level 2	Data Year 2 DCFC	Data Year 3 Level 2	Data Year 3 DCFC
SERVCO	BRENTWOOD	17	3			17	3				
SERVCO	MELVILLE HUB	4	0			4					
	TOTALS	21	3	0	0	21	3	0	0	0	0

KEDLI

Jurisdiction	Location Name	Total Level 2	Total DCFC	The Rate Year - L2 Charger	The Rate Year - DCFC Ch	Data Year 1 Level 2	Data Year 1 DCFC	Data Year 2 Level 2	Data Year 2 DCFC	Data Year 3 Level 2	Data Year 3 DCFC
KEDLI	BELLMORE	29	4			6			4	23	
KEDLI	HEWLETT	13	3		3					13	
KEDLI	ROSLYN	13	2					13	2		
KEDLI	PATCHOGUE	15	2	15			2				
KEDLI	HICKSVILLE	9	2	9	2						
KEDLI	BRIGHTWATERS	9	1			9					1
KEDLI	BRENTWOOD	8	0			8					
KEDLI	SHOREHAM	4	0							4	
KEDLI	GLENWOOD	2	0					2			
KEDLI	HOLTSVILLE	2	0							2	
KEDLI	RIVERHEAD	2	0			2					
	TOTALS	106	14	24	5	25	2	15	6	42	1

KEDNY

Jurisdiction	Location Name	Total Level 2	Total DCFC	The Rate Year - L2 Charger	The Rate Year - DCFC Ch	Data Year 1 Level 2	Data Year 1 DCFC	Data Year 2 Level 2	Data Year 2 DCFC	Data Year 3 Level 2	Data Year 3 DCFC
KEDNY	CANARSIE	97	12	20	2		4	47		30	6
KEDNY	GREENPOINT	60	10	40	5			20			5
KEDNY	STATEN ISLAND	23	5	23					5		
KEDNY	SPRINGFIELD	11	2			11					2
KEDNY	Bay Ridge / Brooklyn	2	0	2							
	TOTALS	193	29	85	7	11	4	67	5	30	13

Date of Request: May 31, 2023 Request No. DPS-375
Due Date: June 12, 2023 NG Request No. NG-385

KeySpan Gas East Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY

KEDNY KEDLI Case 23-G-0225/0226 Data Requests

Request for Information

FROM: Robert Cully

TO: National Grid

SUBJECT: Allocation of Service Company EV Lease Expenses Allocated to KEDNY and KEDLI

Request:

In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

The "Inputs and Assumptions" tab of workbooks "RRP-3 OM Sch 24 Transportation NY23" and "RRP-3 OM Sch 24 Transportation LI23" lists the allocations of Service Company Electric Vehicle (EV) lease expense as 3.69 percent and 2.88 percent, respectively.

1. Explain how the Companies developed these allocation factors.

2. Explain why such allocation methodology is appropriate for allocating the costs of Service Company EVs to KEDNY and KEDLI customers.

Response:

1. Exhibit___(RRP-11), Workpapers to RRP-3, Schedule 24, Workpaper 1 presents total Historic Test Year transportation costs originating from the Service Company and the proportion of that total charged to each receiving company, resulting in allocations of 3.69 percent to KEDNY and 2.88 percent to KEDLI. Please note, however, that these rates are presented for informational purposes only and were not used to allocate electric vehicle lease costs in the Company's Rate Year forecast.

2. As discussed in the response to part 1, these allocation percentages were not used to allocate the lease cost of electric vehicles to KEDNY and KEDLI. Please see Exhibit___(RRP-11), Workpaper to Exhibit RRP-3, Schedule 24, Workpaper 5 for the allocation of Service Company electric vehicle costs to KEDNY and KEDLI. As shown on this workpaper, 100% of the incremental costs for the proposed Service Company electric vehicles are allocated to KEDNY and KEDLI and split between the two companies using the same ratio that total electric vehicle lease expense is split between KEDNY and KEDLI.

After further consideration, the Company submits that the better method to allocate the Service Company electric vehicle lease expense to KEDNY and KEDLI is by using allocation percentages consistent with the approach for allocating costs for the Service Company facilities (Melville and Brentwood) at which the electric vehicles will be located. As discussed in the Company's response to DPS-362, parts 6 and 8, National Grid is installing four level 2 chargers at the Melville facility, which has an allocation of 31% to KEDLI and 54% to KEDNY, and 17 level 2 chargers at the Brentwood facility, with an allocation of 97% to KEDLI and 3% to KEDNY. The Company will allocate the Service Company electric vehicle lease expense proportionally to KEDNY and KEDLI using these ratios, resulting in an allocation of 85% to KEDLI and 15% to KEDNY. The updated allocations will be reflected in the Corrections and Updates filing.

Name of Respondent:
Anthony Pecoriello

Date of Reply:
June 12, 2023

Date of Request: June 2, 2023 Request No. DPS-396
Due Date: June 12, 2023 NG Request No. NG-415

KeySpan Gas East Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY

KEDNY KEDLI Case No. 23-G-0225 & 23-G-0226 Data Requests

Request for Information

FROM: Clifford Tookes

TO: National Grid

SUBJECT: LNG Cybersecurity Upgrade

Request:

In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel, or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

Referring to KEDNY's Gas Infrastructure and Operations Panel (GIOP) testimony, Exhibit___(GIOP-5), as it applies to the Liquefied Natural Gas (LNG) Cyber Security Upgrade:

1. Provide a complete Data Sheet for this line item.

Response:

Please see Attachment 1 for the KEDNY Liquefied Natural Gas (LNG) Cyber Security Upgrade data sheet. A data sheet was not provided in the original filing because the project did not contain funding requests in the Rate Year or Data Years; however, a recent forecast adjustment was required for the project, moving dollars into FY 2025. The updated forecast and data sheet will be reflected in KEDNY's Corrections and Updates testimony.

Name of Respondent: Date of Reply:
Dennis Leyble June 12, 2023

LNG - Cyber Security Upgrade

Type: ☒ **Project** ☐ **Program**	**Category:** ☒ **Capital** ☐ **Both Capital & O&M**	
Investment Code: 5220101837	**Region:** ☒ **KEDNY** ☐ **KEDLI**	
Planning Portfolio: ☐ **Customer Connections** ☐ **Mandated** ☒ **Reliability** ☐ **Non-Infrastructure** ☐ **Indirect**		
Annual Program: ☐	**Estimated Start Date:** 04/01/2024	**Estimated In-Service Date:** 03/31/2025
Sponsor: Leyble, Dennis	**Business Contact:** Gray, Kevin	

Work Description:

Through this project, National Grid's Information Technology (NG IT) organization will enhance cybersecurity protections and support aspects of Operational Technology (OT) in LNG plants with external vendors providing OT support due to their expertise and knowledge of the vaporization and liquefaction control systems in NG LNG plants. This project will be funded under the KEDNY LNG portfolio while managed by NG IT.

This project has two phases:
1.) Enhancements/Remediation - In FY23, third-party vendor, Network + Security Technologies (NST), is assessing each US LNG plant to identify any vulnerabilities and recommend mitigations. Based on NST's assessments in FY23, this phase will implement operational policy improvements, harden OT networks, and complete other work. All US LNG plants will be addressed in this phase, with TSA-critical plants addressed first.
2.) Support - NG IT will provide ongoing support to all US LNG plants to ensure the enhancements implemented during phase one remain intact. NG IT support will start supporting a given plant after enhancements have been implemented in that plant.

Justification Summary:

Necessary cybersecurity enhancements have been identified in LNG plant operational procedures and OT that will be implemented by this project. This project will also enable LNG plants to comply with the new US federal regulations under Transportation Security Administration's (TSA) Security Directive Pipeline-2021-02C for Pipeline Cybersecurity Mitigation Actions, Contingency Planning, and Testing.

Customer Benefits:

The primary customer benefit is the continuous, safe, and reliable supply of natural gas without unplanned outages due to facility shutdowns. Serving as the principal source of "Peak Shaving" for New York City customers, the plant also plays a strategic role in the company's gas supply portfolio. The facility has two (2) LNG storage tanks with a total design capacity of 1.6 BCF of LNG. Greenpoint LNG facility has a design output of 290 MMSCFD of natural gas. LNG peak-shaving facilities operate intermittently, supplying natural gas pipelines during periods of peak need as required.

Securing OT in all US LNG plants will prevent restoration costs from a cyber-attack that renders a single LNG plant inoperable during the winter heating season, reducing customer bill impacts.

Funding Detail

nationalgrid

Cost Breakdown:

$000	FY25	FY26	FY27	FY28
CAPEX	$ 1,202,398	$	$	$
OPEX				
TOTAL	$ 1,202,398	$	$	$

Supplemental Information

Alternatives

Alternative 1: Do nothing – status quo. This alternative is not recommended because it does not mitigate the potential cybersecurity risks in each US LNG plant.

Alternative 2: Hire a third-party vendor to implement cybersecurity enhancements and provide ongoing support without assessing each LNG plant. This alternative has a lower total cost than the selected approach but is not recommended because it is unlikely to comply with TSA Security Directive Pipeline-2021-02C.

Alternative 3: Non-Pipes Alternatives ("NPAs")

NPA Suitability Screening

1. **Is the project construction expected to commence and be completed within 24 months?**
   ☐ Yes ☒ No
2. **For KEDLI or NMPC, is the cost of the project less than $500k? For KEDNY, is the cost of the project less than $750k?**
   ☐ Yes ☒ No
3. **Does the pipes investment affect the critical reliability of the local or broader gas system or respond to a regulatory mandate?**
   ☒ Yes ☐ No

If questions are answered "yes," the project does not qualify for an NPA.

NPA Evaluation

Has this project been reviewed against the Company's NPA Screening and Suitability Criteria?

☒ Yes ☐ No

Please briefly explain the results: Project is not suitable for an NPA as it affects the critical reliability of the gas system and mitigates integrity concerns. No NPA, or combination of NPAs, would remove the need for this project at this time.

CLCPA/GHG Analysis

1. **Is the project/program located in an area designated as a disadvantaged community (DAC)[1]?** ☒ Yes ☐ No
2. **If so, explain how was the routing determined and how will the project impact the surrounding area?** Project is within National Grid's property.
3. **Will the project contribute to GHG emission reductions on the Company's gas network?** ☐ Yes ☒ No
4. **Please explain the project's GHG impact.** No Material Impact. This project is required for reliability and is not anticipated to result in GHG emissions reductions.

Studies/References That Support the Program: TSA Security Directive Pipeline-2021-02C.

Benefits
Reliability Benefits
☐ Supports growth forecast
☒ Addresses supply/capacity constraint or supply diversity needs
☐ Addresses storm/climate change resiliency
Other Customer Benefits:
☐ Improves Customer Satisfaction
☐ Reflects efficiency savings of $___
☐ Reduction in Billing Errors and Re-Bills
☐ Supports implementation of Gas Business Enablement
Safety Benefits:
☐ Enhances response time to Emergency Gas Leaks
☒ Enhances employee safety
☐ Increases automation (reduces human error)
☒ Enhances Public Safety
☐ Reduces Damages potential *(i.e., in the case of mapping system)*
☐ Addresses specific safety initiative: _____
Societal Benefits/Externalities:
☐ Emissions Reduction
☐ Reduces use of Alt. Fuel
☐ Decreases Leak Prone Pipe \Leak Backlog (Type 3 leaks)
☒ Other: Regulatory Compliance

[1] As defined in Environmental Conservation Law Section 75-0101(5), disadvantaged communities are "communities that bear burdens of negative public health effects, environmental pollution, impacts of climate change, and possess certain socioeconomic criteria, or comprise high-concentrations of low- and moderate- income households, as identified pursuant to section 75-0111 of the [Environmental Conservation Law]."